My Learning Draft

Labix

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Contents

1 Combinatorial Commutative Algebra

Pre-req: Commutative Algebra 2, Simplicial Complexes

1.1 Stanley-Reisner Rings

Theorem 1.1.1
Let R be a commutative ring. Let $n \in \mathbb{N}$. There is a bijection

$$\begin{matrix}\text{Abstract Simplicial} \\ \text{Complexes on } \{1,\dots,n\}\end{matrix} \overset{1:1}{\leftrightarrow} \begin{matrix}\text{Squarefree monomial} \\ \text{ideals of } R[x_1,\dots,x_n]\end{matrix}$$

given by sending a simplicial set S to the ideal $\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$.

Definition 1.1.2 (Stanley-Reisner Rings)
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ring of S to be

$$R[S] = \frac{k[x_1,\dots,x_n]}{\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle}$$

The ideal $I_S = \left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$ is called the Stanley-Reisner ideal of S.

Example 1.1.3
Let R be a commutative ring. The Stanley-Reisner ring of the standard n-simplex Δ^n is given by

$$R[\Delta^n] = k[x_1,\dots,x_n]$$

Example 1.1.4
Let R be a commutative ring. The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1,2,3\}) \cup \mathcal{P}(\{1,3,4\}) \cup \mathcal{P}(\{2,4\})$ over $\{1,2,3,4\}$ is given by

$$R[S] = \frac{k[x_1,x_2,x_3,x_4]}{(x_1x_2x_4, x_2x_3x_4)}$$

Proposition 1.1.5
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Then we have

$$I_S = \bigcap_{J\in S} \langle x_i \mid i \notin J \rangle$$

Maps of simplicial sets -> Maps of k-algebra.

Proposition 1.1.6
Let R be a commutative ring. Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Then we have

$$R[\mathcal{K}_1 * \mathcal{K}_2] \cong R[\mathcal{K}_1] \otimes_R R[\mathcal{K}_2]$$

Proposition 1.1.7
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a flag complex, then $R[\mathcal{K}]$ is a Koszul algebra.

1.2 The Hilbert Series of Stanley-Reisner Rings

Theorem 1.2.1 (Stanley)
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S. Let $(h_0, \dots, h_n)$ be the h-vector of S. Then the Hilbert series of $k[S]$ is given by

$$HS_{k[S]}(t) = \sum_{i=0}^{n+1} f_{k-1} \left(\frac{t^2}{1-t^2} \right)^k = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} h_i t^{2i}$$

Example 1.2.2 | Let Δ^n be the standard n-simplex. Then the following are true.

- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by

$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$

- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by

$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} t^{2i}$$

2 Polyhedral Products

2.1 The Homotopy Theory of Polyhedral Products

Definition 2.1.1 (Polyhedral Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces.

- Let $\sigma \in \mathcal{K}$. Define
$$(\mathbf{X},\mathbf{A})^\sigma = \left\{(x_1,\dots,x_n) \in \prod_{i=1}^n X_i \;\middle|\; x_i \in A_i \text{ for } i \notin I\right\}$$
- Define the polyhedral product of $(\mathbf{X},\mathbf{A})$ corresponding to $\mathcal{K}$ to be
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \bigcup_{I\in\mathcal{K}} (\mathbf{X},\mathbf{A})^\sigma$$

Proposition 2.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Then we have
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \operatorname*{colim}_{\sigma\in\text{Cat}(\mathcal{K})} (\mathbf{X},\mathbf{A})^\sigma$$

Proposition 2.1.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ and $(\mathbf{Y},\mathbf{B})$ be n pairs of spaces. Let $f_i : (X_i,A_i) \to (Y_i,B_i)$ be a map of pair for $1 \leq i \leq n$). Then $(f_1,\dots,f_n)$ induces a map
$$(\mathbf{X},\mathbf{A})^\mathcal{K} \to (\mathbf{Y},\mathbf{B})^\mathcal{K}$$

Proposition 2.1.4
Let $\mathcal{K},\mathcal{L}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Let $f_i : (X_i,A_i) \to (Y_i,B_i)$ be a map of pair for $1 \leq i \leq n$). Then we have
$$(\mathbf{X},\mathbf{A})^{\mathcal{K}*\mathcal{L}} = (\mathbf{X},\mathbf{A})^\mathcal{K} \times (\mathbf{X},\mathbf{A})^\mathcal{L}$$

Example 2.1.5
Let $\mathcal{K}$ be a simplicial complex. Then we have
$$\text{cc}(\mathcal{K}) = (I,1)^\mathcal{K}$$

Example 2.1.6
Let $\mathcal{K} = \{\{1,\},\dots,\{n\}\}$. Let $\mathbf{X} = (X_1,\dots,X_n)$ be spaces. Then we have
$$(\mathbf{X}X,*)^\mathcal{K} = X_1 \vee \cdots \vee X_n$$

Definition 2.1.7 (Polyhedral Smash Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ be pairs of spaces.

- Let $\sigma \in \mathcal{K}$. Define
$$(\mathbf{X},\mathbf{A})^{\wedge\sigma} = \left\{(x_1,\dots,x_n) \in \bigwedge_{i=1}^n X_i \;\middle|\; x_i \in A_i \text{ for } i \notin I\right\}$$
- Define the polyhedral smash product of $(\mathbf{X},\mathbf{A})$ corresponding to $\mathcal{K}$ to be
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \bigcup_{I\in\mathcal{K}} (\mathbf{X},\mathbf{A})^{\wedge\sigma}$$

Theorem 2.1.8 (Bahri, Bendersky, Cohen, Gitler)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ be n-pairs of CW complexes. Suppose that the inclusion maps $A_j \to X_j$ are null homotopic for $1 \leq j \leq n$. Then there is a homotopy equivalence

$$(\mathbf{X},\mathbf{A})^{\wedge\mathcal{K}} \simeq \bigvee_{\sigma\in\mathcal{K}} |\mathrm{lk}_{\mathcal{K}}(\sigma)| * (\mathbf{X}_J,\mathbf{A}_J)^{\wedge\sigma}$$

Theorem 2.1.9 (Bahri, Bendersky, Cohen, Gitler)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ be n-pairs of CW complexes. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X},\mathbf{A})^{\mathcal{K}} \simeq \Sigma\left(\bigvee_{J\subseteq\{1,\dots,n\}} (\mathbf{X}_J,\mathbf{A}_J)^{\wedge\mathcal{K}_J}\right)$$

that is natural in $(\mathbf{X},\mathbf{A})$, where $(\mathbf{X}_J,\mathbf{A}_J) = \{(X_j,A_j) \mid j\in J\}$.

Corollary 2.1.10
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ be n-pairs of CW complexes. Suppose that A_i are null homotopic for all i. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X},\mathbf{A})^{\mathcal{K}} \simeq \Sigma\left(\bigvee_{\sigma\in\mathcal{K}} \mathbf{X}^{\wedge\sigma}\right)$$

Corollary 2.1.11
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ be n-pairs of CW complexes. Suppose that X_i are null homotopic for all i. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X},\mathbf{A})^{\mathcal{K}} \simeq \Sigma\left(\bigvee_{J\notin\mathcal{K}} |\mathcal{K}_J| * \mathbf{A}^{\wedge J}\right)$$

Theorem 2.1.12 (Grbic–Theriault, Iriye–Kishimoto)
Let $\mathcal{K}$ be a shifted simplicial complex on $\{1,\dots,n\}$. Let $\mathbf{A}$ be n pointed CW complexes. Then there is a homotopy equivalence

$$(CA,A)^{\mathcal{K}} \simeq \bigvee_{J\notin\mathcal{K}} |\mathcal{K}_J| * \mathbf{A}^{\wedge J}$$

Example 2.1.13
Let $\mathbf{A}$ be n pointed CW complexes. Then there is a homotopy equivalence

$$(CA,A)^{\mathrm{sk}_i\Delta^{n-1}} \simeq \bigvee_{k=i+2}^{n}\left(\bigvee_{1\leq j_1<\cdots<j_k\leq n} (\Sigma^{i+1}A_{j_1}\wedge\cdots\wedge A_{j_k})^{\vee\binom{k-1}{i+1}}\right)$$

2.2 Fat Wedge Filtrations

2.3 Other Results

Proposition 2.3.1
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then we have

$$H^*((\mathbb{CP}^\infty,*)^{\mathcal{K}}) \cong \mathbb{Z}[\mathcal{K}]$$

Proposition 2.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the inclusion map $(\mathbb{CP}^\infty,*)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty)^n$ induces the quotient map

$$\mathbb{Z}[x_1,\dots,x_n] \to \mathbb{Z}[\mathcal{K}]$$

where $\deg(x_i) = 2$.

3 Moment Angle Complexes

3.1 Basic Definitions

Definition 3.1.1 (Moment Angle Complexes)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define the moment angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_\mathcal{K} = (D^2, S^1)^\mathcal{K}$$

Proposition 3.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following diagram is a pullback

$$\begin{tikzcd}
\mathcal{Z}_\mathcal{K} \arrow[r] \arrow[d] & (D^2)^n \subseteq \mathbb{C}^n \arrow[d, "\mu"] \\
\text{cc}(\mathcal{K}) \arrow[r, hook] & I^n
\end{tikzcd}$$

where $D^2 \subseteq \mathbb{C}$ is the closed unit disc and $\mu : (D^2)^n \to I^n$ is the map given by $(z_1,\dots,z_n) \mapsto (|z_1|^2,\dots,|z_n|^2)$.

Corollary 3.1.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then $\mathcal{Z}_\mathcal{K}$ is a T^n-equivariant subspace of $(D^2)^n$.

Example 3.1.4
The following are true.
- $\mathcal{Z}_{\Delta^{n-1}} = (D^2)^n$.
- $\mathcal{Z}_{\partial\Delta^{n-1}} \cong S^{2n-1}$.

Proposition 3.1.5
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$ that is a triangulation of S^m. Then $\mathcal{Z}_\mathcal{K}$ is a compact topological manifold of dimension $m+n+1$.

3.2 Homotopy Theory of Moment Angle Complexes

Proposition 3.2.1
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then there is a factorization

$$\begin{tikzcd}
(\mathbb{CP}^\infty, *)^\mathcal{K} \arrow[rr, hook] \arrow[dr, "h"'] & & (\mathbb{CP}^\infty)^n \\
& ET^n \times_{T^n} \mathcal{Z}_\mathcal{K} \arrow[ur, "p"'] &
\end{tikzcd}$$

where h is a homotopy equivalence and p is a fibration with fiber $\mathcal{Z}_\mathcal{K}$.

Corollary 3.2.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then we have

$$H^*_{T^n}(\mathcal{Z}_\mathcal{K}) \cong \mathbb{Z}[\mathcal{K}]$$

Proposition 3.2.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.
- $\mathcal{Z}_\mathcal{K}$ is 2-connected.
- $\pi_k(\mathcal{Z}_\mathcal{K}) \cong \pi_k((\mathbb{CP}^\infty, *)^\mathcal{K})$ for $k \geq 3$.

3.3 Cohomology of Moment Angle Complexes

Consider the decomposition of D^2 into the following CW complex structure:

- One 0-cell $1 \in D^2$.
- One 1-cell S^1.
- One 2-cell D^2.

Let $n \in \mathbb{N}$. This gives the product complex D^{2n} a CW complex structure, whose cells are a product of 1-cells and 2-cells. We parameterize the cells in the following way. For $J, I \subseteq \{1, \dots, n\}$ with $J \cap I = \emptyset$, we denote $\chi(J, I)$ as the $(|J| + |I|)$-cell corresponding to the cell with the factor of a 1-cell in the jth position for $j \in J$, the factor of a 0-cell in the ith position for $i \in I$, and the factor of a 0-cell for $k \notin I \cup J$.

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a union of cells in D^{2n}, and so is a cellular subcomplex of D^{2n}. Namely, for each $I \in \mathcal{K}$, we have that $\chi(J, I) \subseteq \mathcal{Z}_{\mathcal{K}}$ for all $J \subseteq \{1, \dots, n\} \setminus I$.

Q: How does the cellular cochain split into a bicomplex

Definition 3.3.1 (Bigrading in the Cochain Complex)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define a bigrading in $C^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$ as follows. For each basis element $\chi(J, I)$, define $\deg(\chi(J, I)) = (-|J|, 2(|I| + |J|))$.

Lemma 3.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- The $(2j)$th components for each $j \in \mathbb{N}$ defines a cochain complex $C^{-i,2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$ in i.
- $(C^{-*,*}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}), d_{CW}, 0)$ is a double cochain complex.
- The cellular cochain complex of $\mathcal{Z}_{\mathcal{K}}$ splits into a direct sum
$$C^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \text{Tot}^{\oplus}(C^{-*,*}(\mathcal{Z}_{\mathcal{K}})) = \bigoplus_{j \in \mathbb{N}} C^{-*,2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$$
where
$$C^k(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \bigoplus_{2j-i=k} C^{-i,2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$$

The consequence of the splitting is that cohomology also accquires a splitting
$$H^n(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \bigoplus_{2j-i=n} H^{-i}(C^{-*,2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}))$$

Proposition 3.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- There are natural isomorphisms
$$H^{-p}(C^{-*,2q}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})) \cong H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q} = H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)_{2q}\right)$$
where $\Lambda\left(\bigoplus_{i=1}^n R[\mathcal{K}]u_i\right)$ is the differential graded algebra whose homological grading is given by $\deg(u_i) = -1$, internal grading is given by $\deg(v_i) = 2$ and differential is given by $d(u_i) = v_i$ (and $d(v_i) = 0$ since we think of the DGA as a DGA over the ring $\mathbb{Z}[\mathcal{K}]$).
- There above natural isomorphisms induces natural isomorphisms
$$H^k(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q}$$
- The above natural isomorphisms induce a natural isomorphism of rings
$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong H_*\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)$$

Note that $H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is not isomorphic to $H_{-n}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)$ despite the suggestive notation. This is a mismatch of algebraic objects since the left is an abelian group while the right hand side is a $\mathbb{Z}[\mathcal{K}]$-module. In order to have an isomorphism of abelian groups we must further decompose each homology $\mathbb{Z}[\mathcal{K}]$-modules further into $\mathbb{Z}$-modules by considering $\mathbb{Z}[\mathcal{K}]$ as a $\mathbb{Z}$-module. But this results in a shift of degrees.

Let $K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ be the Koszul complex over $\mathbb{Z}[v_1,\dots,v_n]$. Notice that the cochain complex defined by $K^n = K_n(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ is precisely our differential graded algebra $\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)$. Then the cohomology of $\mathcal{Z}_\mathcal{K}$ acquires an additional description:

$$H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{2q-p=n} \left(H_p\left(K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])\right)\right)_{2q}$$

Finally, since $K_\bullet(v_1,\dots,v_n)$ is a free resolution for $\mathbb{Z}$ as a $\mathbb{Z}[v_1,\dots,v_n]$-module, we know that

$$H_p(K_\bullet(v_1,\dots,v_n;R[\mathcal{K}])) = \text{Tor}_p^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[\mathcal{K}],\mathbb{Z})$$

The Koszul algebra gives the Tor groups the structure of an algebra, and so the cohomology ring acquires the description

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \text{Tor}_*^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[v_1,\dots,v_n],\mathbb{Z})$$

But again note that the nth cohomology group of $\mathcal{Z}_\mathcal{K}$ is not isomorphic to the nth Tor group of $\mathbb{Z}[\mathcal{K}]$ and $\mathbb{Z}$ by virtually the same reason as above.

Corollary 3.3.4

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then there is a homotopy equivalence

$$\Sigma\mathcal{Z}_\mathcal{K} \simeq \bigvee_{J\notin\mathcal{K}} \Sigma^{2+|J|}|\mathcal{K}_J|$$

Corollary 3.3.5

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the above homotopy equivalence descends to a natural isomorphism

$$\bigoplus_{J\subseteq\{1,\dots,n\}} C^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z}) \cong C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$

given by

$$\alpha_L \mapsto \varepsilon(L,J)\chi(J\setminus L, L)$$

where α_L is the cochain basis of Hochster's formula given by the simplex $L\subseteq J$ and

$$\varepsilon(L,J) = \prod_{j\in L}\varepsilon(j,J)$$

and $\varepsilon(j,J) = (-1)^{r-1}$ if j is in the rth position of J.

Proposition 3.3.6

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There above natural isomorphisms induces natural isomorphisms

$$H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) \cong \bigoplus_{\substack{J\subseteq\{1,\dots,n\}\\ |J|=j}} \widetilde{H}^{j-i-1}(\mathcal{K}_J;\mathbb{Z})$$

  where by convention we set $\widetilde{H}^{-1}(\mathcal{K}_\emptyset) = \mathbb{Z}$.
- The above natural isomorphisms induces natural isomorphisms

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

We are missing a description of the cup product in terms of Hochster's formula. It is given by the following definition.

Definition 3.3.7 (Ring Structure in Hochster's Formula)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define a ring structure in $\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_\mathcal{J};\mathbb{Z})$ as follows. Let α_L be the cochain basis element in $C^p(\mathcal{K}_I;\mathbb{Z})$ corresponding to the simplex $L\subseteq I$. The ring structure is induced by the map

$$C^p(\mathcal{K}_I;\mathbb{Z})\otimes_\mathbb{Z} C^q(\mathcal{K}_\mathcal{J};\mathbb{Z})\to C^{p+q+1}(\mathcal{K}_{I\cup J})$$

defined by

$$\alpha_L\otimes\alpha_M\mapsto\begin{cases}\varepsilon(L,I)\varepsilon(M,J)\varepsilon(L\cup M,I\cup J)\alpha_{L\amalg M} & \text{if } I\cap J=\emptyset\\ 0 & \text{otherwise}\end{cases}$$

where $\varepsilon(L,I)=\prod_{j\in L}\varepsilon(j,J)$ and $\varepsilon(j,J)=(-1)^{r-1}$ if j is in the rth position in J.

Proposition 3.3.8
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.
- The ring structure in Hochster's formula is equal to the relative Kunneth theorem for

$$H^*(|\mathcal{K}_I*\mathcal{K}_J|)=H^*(\Sigma(|\mathcal{K}_I|\wedge|\mathcal{K}_J|))=\widetilde{H}^{*-1}(|K_I|\times|K_J|,|K_I\vee K_J|)$$

  via the inclusion $\mathcal{K}_{I\amalg J}\hookrightarrow\mathcal{K}_I*\mathcal{K}_J$.
- The natural isomorphism in the above prp induces a natural isomorphism of rings

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_J;\mathbb{Z})$$

While the cohomology groups can be easily computed using Hochster's formula, the ring structure can be seen more easily using the DGA.

Example 3.3.9
Let $\mathcal{K}$ be the standard polygon with vertices given by $\{1,\dots,5\}$ and edges given by $\{\{1,2\},\dots,\{4,5\},\{1,5\}\}$. We compute that
- $H^0(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\bigoplus_{J\subseteq\{1,\dots,5\}}\widetilde{H}^{0-|J|-1}(\mathcal{K}_\mathcal{J};\mathbb{Z})=\widetilde{H}^{-1-|\emptyset|}(\mathcal{K}_\emptyset;\mathbb{Z})=\mathbb{Z}$.
- $H^1(\mathcal{Z}_\mathcal{K};\mathbb{Z})=H^2(\mathcal{Z}_\mathcal{K};\mathbb{Z})=H^5(\mathcal{Z}_\mathcal{K};\mathbb{Z})=H^6(\mathcal{Z}_\mathcal{K};\mathbb{Z})=0$.

Also we have

$$\begin{aligned}H^3(\mathcal{Z}_\mathcal{K};\mathbb{Z})&=\bigoplus_{J\subseteq\{1,\dots,5\}}\widetilde{H}^{2-|J|}(\mathcal{K}_\mathcal{J};\mathbb{Z})\\&=\widetilde{H}^0(\mathcal{K}_{\{1,3\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{1,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,5\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{3,5\}};\mathbb{Z})\\&=\mathbb{Z}[\alpha_{\{3\}}]\oplus\mathbb{Z}[\alpha_{\{4\}}]\oplus\mathbb{Z}[\alpha_{\{4\}}]\oplus\mathbb{Z}[\alpha_{\{5\}}]\oplus\mathbb{Z}[\alpha_{\{5\}}]\\&=\mathbb{Z}[u_1v_3]\oplus\mathbb{Z}[u_1v_4]\oplus\mathbb{Z}[u_2v_4]\oplus\mathbb{Z}[u_2v_5]\oplus\mathbb{Z}[u_3v_5]\end{aligned}$$

$$\begin{aligned}H^4(\mathcal{Z}_\mathcal{K};\mathbb{Z})&=\bigoplus_{J\subseteq\{1,\dots,5\}}\widetilde{H}^{3-|J|}(\mathcal{K}_\mathcal{J};\mathbb{Z})\\&=\widetilde{H}^0(\mathcal{K}_{\{1,2,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,3,5\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{1,3,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,4,5\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{1,3,5\}};\mathbb{Z})\\&=\mathbb{Z}[\alpha_{\{4\}}]\oplus\mathbb{Z}[\alpha_{\{5\}}]\oplus\mathbb{Z}[\alpha_{\{1\}}]\oplus\mathbb{Z}[\alpha_{\{2\}}]\oplus\mathbb{Z}[\alpha_{\{3\}}]\\&=\mathbb{Z}[-u_1u_2v_4]\oplus\mathbb{Z}[-u_2u_3v_5]\oplus\mathbb{Z}[-u_3u_4v_1]\oplus\mathbb{Z}[-u_4u_5v_2]\oplus\mathbb{Z}[u_1u_5v_3]\end{aligned}$$

and

$$\begin{aligned} H^7(\mathcal{Z}_\mathcal{K};\mathbb{Z}) &= \bigoplus_{J\subseteq\{1,\dots,5\}} \widetilde{H}^{6-|J|}(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \\ &= \widetilde{H}^1(\mathcal{K};\mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{1,2\}}] \\ &= \mathbb{Z}[-u_3u_4u_5v_1v_2] \end{aligned}$$

where wee identified the generators in Hochster's formula to generators in the DGA via the given formulas. Denote the generators of H^3 by $x_1,\dots,x_5$ and denote the generators of H^4 by $y_1,\dots,y_5$ and denote the generator of H^7 by z. Then we have the relations $x_1y_4 = x_2y_2 = -x_3y_5 = x_4y_3 = x_5y_1 = z$, and 0 otherwise. Thus relabelling the generators give

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \frac{\Lambda_\mathbb{Z}[x_1,\dots,x_5,y_1,\dots,y_5]}{(x_iy_i - x_jy_j, x_iy_j, x_ix_j, y_i, y_j \mid 1\leq i\neq j\leq 5)}$$

where $\deg(x_i) = 3$ and $\deg(y_i) = 4$. Notice that this is precisely the cohomology ring of $(S^3\times S^4)^{\#5}$.

There are three descriptions of the cohomology groups:

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q} = \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

For $L\subseteq J$ such that α_L is a cochain basis in the Hochster formula, we have

$$[\varepsilon(L,J)\chi(J\setminus L,L)] \leftrightarrow [\varepsilon(L,J)u_{J\setminus L}v_L] \leftrightarrow \alpha_L$$

Proposition 3.3.10
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- rank$(H^0(C^{-*,0}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = 1$.
- rank$(H^0(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = 0$ for $q\neq 0$.
- rank$(H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z})))$ for $p>q$ or $q>n$.
- rank$(H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z})))$ for $p\geq q>0$ or $q-p>\dim(\mathcal{K})+1$.
- rank$(H^{-1}(C^{-*,4}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- rank$(H^{-(n-\dim(\mathcal{K})+1)}(C^{-*,2n}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = $ rank$(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Proposition 3.3.11
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- rank$(H^0(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = 1$.
- rank$(H^1(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = $ rank$(H^2(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = 0$.
- rank$(H^3(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- rank$(H^{n+\dim(\mathcal{K})+1}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = $ rank$(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Lemma 3.3.12
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1}, then the top cohomology $H^{n+m}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is generated by $[\chi(\{1,\dots,n\}\setminus I, I)]$ for any $I\in\mathcal{K}$ and such that $|I| = n$.

If we use the Koszul complex as the cohomology model for $\mathcal{Z}_\mathcal{K}$, then the top cohomology is generated by $u_{\{1,\dots,n\}\setminus I}v_I$ for any $I\in\mathcal{K}$ such that $|I| = n$.

3.4 Geometry of Moment Angle Complexes

Definition 3.4.1 (Coordinate Subspaces)
Let $I \subseteq \{1, \dots, n\}$. Define the coordinate subspace of I to be

$$L_I = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid z_j = 0 \text{ for all } j \in I\}$$

Definition 3.4.2 (Coordinate Subspace Arrangement)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define the coordinate subspace arrangement of $\mathcal{K}$ to be

$$U(\mathcal{K}) = \mathbb{C}^n \setminus \bigcup_{\sigma \in \mathcal{K}} L_\sigma$$

Proposition 3.4.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then we have $U(\mathcal{K}) = (\mathbb{C}, \mathbb{C}^\times)^{\mathcal{K}}$.

Theorem 3.4.4
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. The map $(\mathbb{C}, \mathbb{C}^\times) \to (D^2, S^1)$ induces a T^n-equivariant deformation retract

$$U(\mathcal{K}) \simeq \mathcal{Z}_{\mathcal{K}}$$

4 Geometric Structures of Moment Angle Complexes

4.1 Moment Angle Polytopes and Intersection of Quadrics

Let $P \subseteq \mathbb{R}^n$ be a polytope defined by

$$P = \{x \in \mathbb{R}^n \mid \langle a_i, x\rangle + b_i \geq 0 \text{ for } 1 \leq i \leq m\}$$

such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Let

$$A = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \in M_{m\times n}(\mathbb{R}) \quad \text{and} \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \mathbb{R}^m$$

Recall that

$$P = \{x \in \mathbb{R}^n \mid Ax + b \geq 0\}$$

Definition 4.1.1 (Moment Angle Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Let $i_P : \mathbb{R}^n \to \mathbb{R}^m$ be the map $i_P(x) = Ax - b$. Let $\mu : \mathbb{C}^m \to \mathbb{R}^m$ be the map $\mu(z_1, \dots, z_m) = (|z_1|^2, \dots, |z_m|^2)$. Define the moment angle polytope $\mathcal{Z}_{A,b}$ to be the pullback

$$\begin{tikzcd} \mathcal{Z}_{A,b} \arrow[r] \arrow[d] & \mathbb{C}^n m \arrow[d, "\mu"] \\ P \arrow[r, hook, "i_P"] & \mathbb{R}^m \end{tikzcd}$$

Proposition 4.1.2
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then there is a homeomorphism

$$\mathcal{Z}_{A,b} \cong \left\{ z = (z_1, \dots, z_m) \in \mathbb{C}^m \;\middle|\; \Gamma \begin{pmatrix} |z_1|^2 \\ \vdots \\ |z_m|^2 \end{pmatrix} = \Gamma b \right\}$$

where $\Gamma = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_{m-n} \end{pmatrix}$ and $\gamma_1, \dots, \gamma_{m-n}$ is a basis for $\ker(A^T)$.

This makes $Z_{A,b}$ an intersection of quadrics in $\mathbb{R}$, as well as an intersection of Hermitian quadrics (Hermitian since $|z_i|^2 = z_i\overline{z_i}$).

Definition 4.1.3 (Intersection of Quadrics)
Let $\Gamma \in M_{(m-n)\times n}(\mathbb{R})$ be a matrix. Let $\delta \in \mathbb{R}^{m-n}$. Define the intersection of quadrics associated to (Γ, δ) by

$$\mathcal{Z}_{\Gamma,\delta} = \left\{ z = (z_1, \dots, z_m) \in \mathbb{C}^m \;\middle|\; \Gamma \begin{pmatrix} |z_1|^2 \\ \vdots \\ |z_m|^2 \end{pmatrix} = \delta \right\}$$

Lemma 4.1.4
Let $\Gamma \in M_{(m-n)\times n}(\mathbb{R})$ be a matrix. Let $\delta \in \mathbb{R}^{m-n}$. Then we have

$$\frac{\mathcal{Z}_{\Gamma,\delta}}{T^m} \cong \left\{ y = (y_1, \dots, y_m) \in \mathbb{R}^m \;\middle|\; \Gamma y = \delta \right\}$$

Definition 4.1.5 (Associated Polytope)
Let $\Gamma \in M_{(m-n)\times n}(\mathbb{R})$ be a matrix. Let $\delta \in \mathbb{R}^{m-n}$. Define the associated polytope of $\mathcal{Z}_{\Gamma,\delta}$ to be $\mathcal{Z}_{A,b}$ where

- The columns $a_1, \dots, a_m$ of A form a basis for $\ker(\Gamma^T)$.
- $b \in \mathbb{R}^m$ is such that $\Gamma b = \delta$.

Theorem 4.1.6
There is a one to one correspondence

$$\frac{\left\{\mathcal{Z}_{A,b} \;\middle|\; \substack{A\in M_{m\times n}(\mathbb{R}),b\in\mathbb{R}^m \text{ such that} \\ \text{the columns } a_1,\dots,a_m \text{ of } A \text{ span } \mathbb{R}^n}\right\}}{\text{Isomorphism of } \mathbb{R}^n} \overset{1:1}{\longleftrightarrow} \frac{\left\{\mathcal{Z}_{\Gamma,\delta} \;\middle|\; \substack{\Gamma\in M_{(m-n)\times n}(\mathbb{R}),\delta\in\mathbb{R}^{m-n} \text{ such that} \\ \text{the columns } \gamma_1,\dots,\gamma_m \text{ of } \Gamma \text{ span } \mathbb{R}^{m-n}}\right\}}{\text{Linear isomorphism of } \mathbb{R}^{m-n}}$$

Proposition 4.1.7
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then P is simple if and only if $\mathcal{Z}_{A,b}$ is a smooth manifold of real dimension $m + n$.

4.2 Relation to Moment Angle Complexes

Proposition 4.2.1
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then there is a homotopy equivalence

$$\mathcal{Z}_{\mathcal{K}_P} \simeq \mathcal{Z}_{A,b}$$

Proposition 4.2.2
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. If P is simple, then there is a T^m-equivariant homeomorphism

$$\mathcal{Z}_{\mathcal{K}_P} \cong \mathcal{Z}_{A,b}$$

There is a canonical choice of intersection of quadrics in this case.

Proposition 4.2.3
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Suppose that P is simple. Then there is a T^m-equivariant diffeomorphism

$$\mathcal{Z}_{\mathcal{K}_P} \cong \left\{ z = (z_1, \dots, z_m) \in \mathbb{C}^m \;\middle|\; \sum_{k=1}^m |z_k|^2 = 1, \sum_{k=1}^m g_k |z_k|^2 = 0 \right\}$$

where $g_1, \dots, g_m \in \mathbb{R}^{m-n-1}$ is a combinatorial gale diagram of P^*.

Theorem 4.2.4 (de Medrano)
Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Suppose that $\mathcal{Z}_{\mathcal{K}_P}$ is a non-empty and non-degenerate intersection of 3 quadrics. Then exactly one of the following hold.

- P is combinatorially equivalent to a product of three simplicies, and there is a diffeomorphism
$$\mathcal{Z}_{\mathcal{K}_P} \cong S^{n_1} \times S^{n_2} \times S^{n_3}$$
where n_1, n_2, n_3 are odd.
- There is a diffeomorphism
$$\mathcal{Z}_{\mathcal{K}_P} \cong \#_{k=3}^{n+1} (S^k \times S^{2n+3-k})^{\# q_k}$$
for some q_k determined by the Gale diagram of P.

5 Real Moment Angle Complexes

Definition 5.0.1 (Real Moment Angle Complexes)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define the real moment angle complex of $\mathcal{K}$ to be

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} = (D^1, S^0)^{\mathcal{K}}$$

Proposition 5.0.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following diagram is a pullback

$$\begin{tikzcd}
\mathbb{R}\mathcal{Z}_{\mathcal{K}} \arrow[r] \arrow[d] & (D^1)^n \subseteq \mathbb{C}^n \arrow[d, "\mu"] \\
\operatorname{cc}(\mathcal{K}) \arrow[r, hook] & I^n
\end{tikzcd}$$

where $D^1 \subseteq \mathbb{R}$ is the closed unit disc and $\mu : (D^2)^n \to I^n$ is the map given by $(x_1, \dots, x_n) \mapsto (x_1^2, \dots, x_n^2)$.

Example 5.0.3
We have

$$\mathcal{Z}_{\partial\Delta^{n-1}} \cong S^{n-1}$$

Proposition 5.0.4
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$ that is a triangulation of S^m. Then $\mathcal{Z}_{\mathcal{K}}$ is a compact topological manifold of dimension $m+1$.

Example 5.0.5
Let $\mathcal{K}$ be the boundary of an n-gon. Then there is a homeomorphism

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} \cong \Sigma_{1+(n-4)2^{n-3}}$$

Corollary 5.0.6
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is a homotopy equivalence

$$\Sigma\mathbb{R}\mathcal{Z}_{\mathcal{K}} \simeq \bigvee_{J \notin \mathcal{K}} \Sigma^2 |\mathcal{K}_J|$$

5.1 The Fat Wedge Filtration of Real Moment Angle Complexes

Recall that cubical complex of a simplicial complex. We know that

$$(D^1, S^0)^{\sigma} = \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

and

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} = \bigcup_{\sigma \in \mathcal{K}} \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

We consider the fat wedge filtration of $\mathbb{R}\mathcal{Z}_{\mathcal{K}}$. We note that at the ith step, we have

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}}^i = \bigcup_{I \subseteq \{1, \dots, n\}, |I| = i} \mathbb{R}\mathcal{Z}_{\mathcal{K}_I}$$

Proposition 5.1.1
Let $n \in \mathbb{N}^+$. Let $\mathcal{K}$ be a simplicial complex on n. Let $J \subseteq \mathcal{K}$. Then we have

$$(|\text{Cone}(\text{Bd}(\mathcal{K}_J))|, |\text{Bd}(\mathcal{K}_J)|) \cong (\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}, \mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1})$$

Proof
Let $i_c : |\Delta^n| \to I^{n+1}$ denote the cubical embedding. For each $J \subseteq \{1, \dots, n\}$, the embedding restricts to a map $i_c^J : |\mathcal{K}_J| \to I^J$ where $I^J = (I, 0)^J$ is the moment angle complex of I. For each $\mu \subseteq \tau \subseteq J$, denote $C^J_{\mu\subseteq\tau}$ the $C_{\mu\subseteq\tau}$ face of I^J (so $C^J_{\mu\subseteq\tau}$ still lives inside I^J but some coordinates are 0). We compute that

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1} = \bigcup_{A\subseteq J, |A|=|J|-1} \bigcup_{\sigma\in\mathcal{K}_A} \bigcup_{\substack{\mu\subseteq\tau\subseteq A \\ \tau\setminus\mu=\sigma}} C^A_{\sigma\subseteq\tau} = |\text{Bd}(\mathcal{K}_J)|$$

and similarly

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J} = \bigcup_{\sigma\in\mathcal{K}_J} \bigcup_{\substack{\mu\subseteq\tau\subseteq J \\ \tau\setminus\mu=\sigma}} C^J_{\mu\subseteq\tau} = |\text{Cone}(\text{Bd}(\mathcal{K}_J))|$$

■

Let us verify this for the case $\mathcal{K} = \partial\Delta^2$. Pick $I = \{1\}$. Let $i_c : |\mathcal{K}| \to I^n$ be the map that is the cubical embedding. Now the relative version of the identity

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K}) \\ \tau\setminus\mu=\sigma}} C_{\mu\subseteq\tau}$$

says that if $\sigma \in \mathcal{K}_I$, then

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K}) \\ \tau\setminus\mu=\sigma}} C^I_{\mu\subseteq\tau}$$

where we consider $(D^1, S^0)^\sigma$ as a subspace of $\mathcal{Z}_\mathcal{K}$ by considering $\mathcal{Z}_{\mathcal{K}_I}$ as a subspace of $\mathcal{Z}_\mathcal{K}$, and on the right $C^I_{\mu\subseteq\tau}$ is the usual meaning but restricted to the cube $(D^1)^I$ of I^3.

Restricting this map to $\mathcal{K}_I$, this the map sending $\{1\}$ to $C^I_{\{1\},\{1\}}$ which is the point $-1 \in C^I_{\{1\},\{1\}} = D^1$, but is the point $(-1, 1, 1)$ in I^3.

6 Research Notes

The Foundation

Moment angle complexes are T^n-equivariant objects defined using combinatorial data. More specifically, choose a simplicial complex $\mathcal{K}$ on $\{1, \dots, n\}$. For each simplex $\sigma \in \mathcal{K}$, define the space

$$(D^2, S^1)^\sigma = \prod_{i=1}^{n} Y_i \quad \text{where} \quad Y_i = \begin{cases} D^2 & \text{if } i \in \sigma \\ S^1 & \text{otherwise} \end{cases}$$

In other words, $(D^2, S^1)^\sigma$ is a product of D^2 and S^1 depending on which vertices of $\mathcal{K}$ lie in σ. For any inclusion of simplicies $\sigma \subseteq \tau$, there is an inclusion map $(D^2, S^1)^\sigma \hookrightarrow (D^2, S^1)^\tau$. Then the definition of the moment angle complex $\mathcal{Z}_\mathcal{K}$ of $\mathcal{K}$ is just the gluing of products of D^2 and S^1 along these inclusions, or

$$\mathcal{Z}_\mathcal{K} = \bigcup_{\sigma \in \mathcal{K}} (D^2, S^1)^\sigma$$

Being combinatorial in nature, they form a testing ground for conjectures in topology. Moreover, in good cases (eg $\mathcal{K}$ is a triangulation of S^m), the moment angle complex $\mathcal{Z}_\mathcal{K}$ has the structure of a smooth manifold and they form a testing ground for conjectures in geometry.

These topological objects are also interesting because they lie at the intersection of a number of active fields. Other than moment angle complexes, there are other objects in topology and geometry that are defined in terms of combinatorial objects

$$\begin{array}{ccc} \mathcal{Z}_\mathcal{K} & \mathcal{Z}_P & V_P \\ & & \\ \text{(Topology)} & \text{(Symplectic Geometry)} & \text{(Algebraic Geometry)} \end{array}$$

Moment angle polytopes $\mathcal{Z}_P$ are intersection of Hermitian quadrics that define non-Kahler complex manifolds, also makes for a testing ground for theories in symplectic and complex geometry.

Projective toric varieties V_P are combinatorial objects in algebraic geometry formed by certain types of polytopes that are also, explicit enough to form a testing ground for conjectures in algebraic geometry. Choosing our polytopes carefully connects the three objects into a unified theory.

$$\begin{tikzcd}
U(\mathcal{K}) \arrow[r, "\cong_{T^m}"] \arrow[d, "\simeq_{T^m}"] & U(\Sigma_P) \arrow[r, "T^m\text{-Orbit}"] \arrow[d, "\simeq_{T^m}"] & U(\Sigma_P)/T^m \arrow[r, "\cong"] \arrow[d, "\cong\text{diffeo}"] & V_P \\
\mathcal{Z}_\mathcal{K} \arrow[r, "\cong_{T^m}"] & \mathcal{Z}_P \arrow[r, "T^{m-n}\text{-Orbit}"] & \mathcal{Z}_P/T^{m-n} &
\end{tikzcd}$$

(Topology) (Symplectic Geometry) (Algebraic Geometry)

In fact, the first connection between symplectic geometry and combinatorics came from the following theorem.

> **Theorem 6.0.1** (The Atiyah–Guillemin–Sternberg convexity theorem (1982))
> Let (M, ω) be a compact connected symplectic manifold. Let $T^k \times M \to M$ be a Hamiltonian torus action. Then $\mu(M)$ is a convex polytope in $\text{Lie}(T^k)^*$ given by
> $$\mu(M) = \text{Conv}(\text{Fix}_{T^k}(M))$$

Delzant in 1988 proved that if the action is more over effective and M, then these polytopes completely determine the manifold structure. This motivates the definition of quasitoric manifolds: Smooth manifolds M^{2n} with a torus action T^n that locally looks like the standard action, together with a homeomorphism of the orbit space $M/T^n \cong P$ with a polytope. In 2008, the following conjecture is made.

Conjecture 6.0.2 (Masuda–Suh (2008))
If M and N are quasitoric manifolds such that

$$H^*(M) \cong H^*(N)$$

Then M and N are homeomorphic (or diffeomorphic).

We are interested in the cohomological rigidty of moment angle complexes instead. The first observation in this direction is that Hochster's formula shows that the cohomology of a moment angle complex is completely determined by the simplicial complex:

Theorem 6.0.3 (Hochster's Formula)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is an isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} H^*(\mathcal{K}_J)$$

where
- $\mathcal{K}_J$ is the full subcomplex of $\mathcal{K}$ with the vertices in J.
- The additive isomorphism is given by

$$H^n(\mathcal{Z}_{\mathcal{K}}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} H^{n-|J|-1}(\mathcal{K}_J)$$

- The ring structure is given the map $H^i(\mathcal{K}_I) \otimes H^j(\mathcal{K}_J) \to H^{i+j+1}(\mathcal{K}_{I \cup H_J})$ which sends products of generators to generators if $I \cap J = \emptyset$ and 0 otherwise.

The second observation is that certain classes of simplicial complexes have the geometry of connected sums of products of spheres:

Theorem 6.0.4
Let P be a stacked polytope. Let $n = \dim(P)$ and $m = |V(P)|$. Then there is a diffeomorphism

$$\mathcal{Z}_{\partial P} \cong \#_{k=3}^{m-n+1} (S^k \times S^{m+n-k})^{(k-2)\binom{m-n}{k-1}}$$

Among the class of topological spaces, spheres are homologically and homotopically the easiest to understand. It is natural to ask the question in what conditions does a moment angle complex has its cohomology only involving spheres, and in this case is it possible to lift it to a homotopy equivalence, a homeomorphism or a diffeomorphism. Currently my work focuses on moment angle complexes coming from simplicial polytopes of dimension 4 with 8 vertices.

What is Known in the Simplicial 4-Polytopes and 8-Vertices Cases

There are 39 simplicial complexes in dimension 4 with 8 vertices (no ghost vertices). Of which 37 of them come from the boundary of a simplicial 4-polytope with 8 vertices (polytopal spheres). These are classified in [An Enumeration of Simplicial 4-Polytopes with 8-Vertices] by Grunbaum and Sreedharan, and whose list was later corrected by my supervisor.

P_1^8 is a stacked polytope where:

- The starting simplex is 12356.
- Vertex 4 is stacked on the facet 1235.
- Vertex 7 is stacked on the facet 1236.
- Vertex 8 is stacked on the facet 1256.

P_2^8 is a stacked polytope where:

- The starting simplex is 12356.

- Vertex 4 is stacked on the facet 1235.
- Vertex 7 is stacked on the facet 1236.
- Vertex 8 is stacked on the facet 1245.

P_3^8 is a stacked polytope where:

- The starting simplex is 12345.
- Vertex 6 is stacked on the facet 1235.
- Vertex 7 is stacked on the facet 1236.
- Vertex 8 is stacked on the facet 1234.

For a stacked polytope P, we have a theorem

Theorem 6.0.5
Let P be a stacked polytope. Let $n = \dim(P)$ and $m = |V(P)|$. Then there is a diffeomorphism

$$\mathcal{Z}_{\partial P} \cong \#_{k=3}^{m-n+1}(S^k \times S^{m+n-k})^{(k-2)\binom{m-n}{k-1}}$$

Hence we conclude that

$$\mathcal{Z}_{\partial P_1^8} \cong \mathcal{Z}_{\partial P_2^8} \cong \mathcal{Z}_{\partial P_3^8} \cong (S^3 \times S^9)^{\#6} \# (S^4 \times S^8)^{\#8} \# (S^5 \times S^7)^{\#3}$$

Also P_{35}^8, P_{36}^8 and P_{37}^8 are neighbourly polytopes (one such is cyclic), and we have the following theorem.

Theorem 6.0.6 (Membrillo Solis, Theriault (2026))
Let $\mathcal{K}$ be a simplicial complex such that the following are true.

- $\mathcal{K}$ is a triangulation of S^{2n+1}.
- $\mathcal{K}$ is neighbourly.

Then there is a homotopy equivalence

$$(D^k, S^{k-1})^{\mathcal{K}} \simeq \#_{i=1}^{j} \left(S^{t_i} \times S^{s_i}\right)$$

for some $j \in \mathbb{N}^+$ and some $t_i, s_i \in \mathbb{N}^+$ for all $k \geq 2$.

So we have

$$\mathcal{Z}_{\partial P_{35}^8} \simeq \mathcal{Z}_{\partial P_{36}^8} \simeq \mathcal{Z}_{\partial P_{37}^8} \simeq (S^5 \times S^7)^{\#16} \# (S^6 \times S^6)^{\#15}$$

Also P_{34}^8 is a cross polytope (It is given by $P_{34}^8 = S^0 * S^0 * S^0 * S^0$). A direct computation shows that

$$H^*(\mathcal{Z}_{\partial P_{34}^8}) \cong H^*(S^3 \times S^3 \times S^3 \times S^3)$$

and the following theorem lifts this isomorphism into a homeomorphism.

Theorem 6.0.7 (Fan, Cheng, Ma, Wang (2014))
Let $\mathcal{K}$ be a simplicial complex such that $\mathcal{K}$ is a triangulation of S^n for $n \geq 2$. Suppose that

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong H^*\left(\#_{i=1}^{j}\left(\prod_{k=1}^{t_i} S^{d_k}\right)\right)$$

Then the following are true.

- If $t_i = n+1$ for some i, then $j = 1$ and there is a homeomorphism
$$\mathcal{Z}_{\mathcal{K}} \cong \prod_{i=1}^{n+1} S^3$$
- We have
$$\left|\left\{ i \in \mathbb{N} \;\middle|\; t_i \geq \left\lfloor \frac{n}{2} \right\rfloor + 2 \right\}\right| \leq 1$$

(MOMENT-ANGLE MANIFOLDS AND CONNECTED SUMS OF SPHERE PRODUCTS)

For the remainder of the simplicial 4-polytopes with 8 vertices, Oganisian and Panov gave an equivalent characterization for those that have the cohomology of a connected sum of products of spheres.

Theorem 6.0.8 ((Oganisian, Panov (2024)))
Let $\mathcal{K}$ be a simplicial complex such that $\mathcal{K}$ is a triangulation of S^3. Then

$$H^*(\mathcal{Z}_\mathcal{K}) \cong H^*\left(\#_{i=1}^j \left(\prod_{k=1}^{t_i} S^{d_k}\right)\right)$$

if and only if exact;y one of the following are true.
- $\mathcal{K} = S^0 * S^0 * S^0 * S^0$.
- $G(\mathcal{K}^1)$ is chordal.
- $\mathcal{K}^1$ has exactly two missing edges with distinct vertices (forming a square).

The same paper by Fan, Cheng, Ma, Wang also shows:

$$H^*(\mathcal{Z}_{\partial P^8_{27}}) \cong H^*(\mathcal{Z}_{\partial P^8_{28}}) \cong H^*(\mathcal{Z}_{\partial P^8_{29}}) \cong H^*\left((S^3\times S^3\times S^6)\#(S^5\times S^7)^{\#8}\#(S^6\times S^6)^{\#8}\right)$$

The paper [Iriye (2016)] ON THE MOMENT-ANGLE MANIFOLD CONSTRUCTED BY FAN, CHEN, MA AND WANG proves that there is a diffeomorphism

$$\mathcal{Z}_{\partial P^8_{28}} \cong (S^3\times S^3\times S^6)\#(S^5\times S^7)^{\#8}\#(S^6\times S^6)^{\#8}$$

Summary:

- The diffeomorphism types of the moment angle complexes of $P^8_1, P^8_2, P^8_3, P^8_{28}$ are understood (they are the stacked polytopes of dim 4 and vert 8 together with the example given by FCMW).
- The homeomorphism type of the moment angle complexe of P^8_{34} is understood (it is the cross polytope of dim 4).
- The homotopy types of the moment angle complexes of $P^8_{35}, P^8_{36}, P^8_{37}$ are understood (they are the neighbourly polytopes of dim 4 and vert 8).
- The conditions for the cohomology type of moment angle complexes of simplicial complexes of dimension 3 (and hence simplicial 4-polytopes) to be the connected sum of products of spheres is understood. By direct computation and the assistance of python, we conclude that the moment angle complexes of $P^8_7, P^8_{16}, P^8_{17}, P^8_{22}, P^8_{26}$ do not have the cohomology of a connected sum of products of spheres, while all the others do.

Techniques and Research Directions

Going forward, there are a number of possible directions that I will focus my work on, in terms of the cohomology, the topology and the rigidity of moment angle complexes.

Analyzing the Topology and Geometry

There are retractions $\mathcal{Z}_{\mathcal{K}_J} \hookrightarrow \mathcal{Z}_\mathcal{K} \to \mathcal{Z}_{\mathcal{K}_J}$ where $J \subseteq \{1,\dots,n\}$ induced by retracting (D^2,S^1) to (S^1,S^1) by mapping D^2 to the base point, and these are typically easier to compute. In particular, the retraction may factor through the $(n-1)$-skeleton, which may reveal information about the topology of the moment angle complex. In [Membrilo Solis, Theriault (2026)] and [Iriye (2018)], certain $\mathcal{Z}_{\mathcal{K}_J}$ are wedges of spheres which are useful to knowing the topology because the remainder of $\mathcal{Z}_{\mathcal{K}_J}$ in $\mathcal{Z}_\mathcal{K}$ may be understood by attaching maps of spheres to wedges of spheres, and if this is the case we may use Hilton Milnor theorem to analyze the components.

In the general case, [BBCG] only decomposes the suspension of $\mathcal{Z}_\mathcal{K}$ as a wedge of spheres:

$$\Sigma\mathcal{Z}_\mathcal{K} \simeq \bigvee_{J\notin\mathcal{K}} \Sigma^{2+|J|}|\mathcal{K}_J|$$

And work was done to desuspend this decomposition [Iriye, Kishimoto], the case for a shifted complex or disjoint points.

Motivated by [Membrillo Solis, Theriault] and [Iriye], I will focus my work in the following way.

1. Analyze the moment angle complexes coming from subcomplexes of $\mathcal{K}$. It is expected that have the homotopy type of a wedge of spheres.
2. Prove that the CW skeleton of $\mathcal{Z}_\mathcal{K}$ has the homotopy type of a wedge of spheres, and analyze the remaining attaching maps to identify the homotopy type of $\mathcal{Z}_\mathcal{K}$.
3. Use the theory of fiber bundles and cobordism following [Iriye] to conclude that the diffeomorphism type of $\mathcal{Z}_\mathcal{K}$.

This will be the main goal of my next year and will be done first to a handful of examples, and then I will generalize the procedure to a larger class of polytopes.

Obstructions to Rigidity

Massey products cannot be detected in cohomological isomorphisms so it is a prime method to showing two spaces are not homeomorphic. There are combinatorial descriptions [Denham, Suciu (2005)] for whitehead products to be non-trivial using the underlying graph of simplicial complexes. More generally the same can be said for A_∞ structures on the cohomology ring.

Analyzing the Cohomology

If $\mathcal{K}$ is a simplicial complex that is a triangulation of S^n. Then the following are true.

- $\mathcal{K}_{[m]\setminus J}$ is a deformation retract of $\mathcal{K} \setminus \mathcal{K}_J$.
- $|\mathcal{K}_J|$ is a compact locally contractible subspace of $|\mathcal{K}|$.
- By Alexander duality, we have

$$\widetilde{H}^i(\mathcal{K}_J) \cong \widetilde{H}_{n-i-1}(S^n \setminus \mathcal{K}_J) \cong \widetilde{H}_{n-i-1}(\mathcal{K} \setminus \mathcal{K}_J) \cong \widetilde{H}_{n-i-1}(\mathcal{K}_{[m]\setminus J})$$

I also explicitly computed

$$H^*(\mathcal{Z}_{\partial P_7^8}) = H^*((S^3 \times S^9)^{\#3} \#(S^4 \times S^8)^{\#3} \#(S^5 \times S^7) \# X)$$

where

$$H^*(X) = \frac{\mathbb{Z}\langle a_1, a_2, b_1, b_2, c, d_1, d_2, e, f_1, f_2, g_1, g_2, h\rangle}{I}$$

and I is the ideal generated by the following relations:

- $a_1 a_2 = d_2$.
- $a_i b_i = e$.
- $a_i c = f_i$.
- $a_1 d_1 = g_2, a_2 d_2 = g_1$.
- $b_i c = g_i$.
- Poincare relations $a_i g_i = b_i f_i = ce = d_1 d_2 = h$.
- $z^2 = 0$ for all generators z.

and $\deg(a_i) = 3$, $\deg(b_i) = 4$, $\deg(c) = 5$, $\deg(d_i) = 6$, $\deg(e) = 7$, $\deg(f_i) = 8$, $\deg(h) = 12$ (13 generators and 28 relations). It is interesting to find a model for X and to see find the homotopy type of $\mathcal{Z}_\mathcal{K}$ in these odd cases, and I aim to use Macaulay2 to simply the above expression and the cohomology rings of the other odd cases.

Some Computations

The following computation is based on my supervisor's paper.

Example 6.0.9

Let $\mathcal{K}$ be the pentagon. We aim the homotopy type of $\mathcal{Z}_\mathcal{K}$. Using Hochster's formula, we deduce that

$$H^*(\mathcal{Z}_\mathcal{K}) \cong H^*\left((S^3 \times S^4)^{\#5}\right)$$

Since $\mathcal{Z}_\mathcal{K}$ is simply connected, we use Hatcher 4C.1 to conclude that $\mathcal{Z}_\mathcal{K}$ has one 0-cell, five 3-cells, five 4-cells and one 7-cell. There is only one way that the five 3-cells can attached to the 0-cell, hence the three skeleton is just $\bigvee_{i=1}^5 S^3$. For a four cell, the attaching map takes the form $S^3 \to \bigvee_{i=1}^5 S^3$. Using Hilton Milnor theorem, we see that the only possible attaching map is by pushing out along a map $S^3 \to S^3$ with $S^3 \to *$. Since the cohomology of $\mathcal{Z}_\mathcal{K}$ has no torsion, this map must be homotopic to the identity, and hence the four skeleton is given by $\overline{\mathcal{Z}_\mathcal{K}} = \bigvee_{i=1}^5(S^3 \vee S^4)$. For the top cell, we also use Hilton Milnor theorem to get

$$\pi_6\left(\overline{\mathcal{Z}_\mathcal{K}}\right) \cong \pi_6(S^3)^5 \times \pi_6(S^4) \times \pi_6(S^5)^{10} \times \pi_6(S^6)^{25}$$

where the factors into the copies of S^3 and S^4 are given by the inclusions $S^6 \to S^{3,4} \hookrightarrow \overline{\mathcal{Z}_\mathcal{K}}$, the third factor is given by $S^6 \to S^5 \to S^3 \vee S^3 \hookrightarrow \overline{\mathcal{Z}_\mathcal{K}}$ the whitehead product and the last map is given by an element of $\pi_6(S^6)$ composed with the Whitehead product $S^6 \to S^6 \to S^3 \vee S^4 \hookrightarrow \overline{\mathcal{Z}_\mathcal{K}}$.

For the element in $\pi_6(S^5)$, note that the composition of $S^6 \to S^5 \to S^3 \vee S^3 \hookrightarrow \overline{\mathcal{Z}_\mathcal{K}}$ with the inclusion $\overline{\mathcal{Z}_\mathcal{K}} \to \mathcal{Z}_\mathcal{K}$ is null homotopic since it is a pushout. Let $J \subseteq \{1, \dots, 5\}$ be a subset of four elements. Then we have $\mathcal{Z}_{\mathcal{K}_J} = S^3 \vee S^3 \vee S^3 \vee S^4 \vee S^4$. There are 5 choices of J that contains all possible permutations of the choices of two S^3 that make up the whitehead product $S^5 \to S^3 \vee S^3$ in $\overline{\mathcal{Z}_\mathcal{K}}$. On the other hand, the inclusion $S^1 \hookrightarrow D^2$ and the projection of D^2 to $1 \in S^1$ induces retractions

$$\mathcal{Z}_{\mathcal{K}_J} \hookrightarrow \mathcal{Z}_\mathcal{K} \to \mathcal{Z}_{\mathcal{K}_J}$$

Since $\mathcal{Z}_{\mathcal{K}_J}$ has cells of at most dimension 4, this retraction factors through $\overline{\mathcal{Z}_\mathcal{K}}$. Since $S^6 \to \overline{\mathcal{Z}_\mathcal{K}} \to \mathcal{Z}_\mathcal{K}$ is null homotopic, post composition with the retraction $\mathcal{Z}_\mathcal{K} \to \mathcal{Z}_{\mathcal{K}_J}$ is also null homotopic. Since $\mathcal{Z}_{\mathcal{K}_J} \hookrightarrow \overline{\mathcal{Z}_\mathcal{K}} \to \mathcal{Z}_{\mathcal{K}_J}$ is the identity, together with null homotopy this implies that $S^6 \to S^5 \to S^3 \vee S^3 \hookrightarrow \mathcal{Z}_{\mathcal{K}_J}$ is null homotopic. But $S^5 \to S^3 \vee S^3$ is the Whitehead map, hence $S^6 \to S^5$ is null homotopic.

The case for the elements in $\pi_6(S^3)$ and $\pi_6(S^4)$ is similar but we use the subcomplexes $J \subseteq \{1, \dots, 5\}$ with $\mathcal{K}_J$ being two disjoint points for the first case, and the subcomplex $J \subseteq \{1, \dots, 5\}$ with $|J| = 3$.

For the element in $\pi_6(S^6)$, follow a similar argument in Hatcher 4C.2 to convert the map $S^6 \to S^6 \to S^3 \vee S^4 \to \mathcal{Z}_\mathcal{K}$ into reading it off using cup products. Then since the cohomology of $\mathcal{Z}_\mathcal{K}$ is completely identified, we see that the coefficient of this map is ± 1 in 5 cases and 0 in the other 20 cases. (Notice that the argument of $\pi_6(S^5)$ does not work for $\pi_6(S^6)$ because there are not enough subcomplexes that give all possible choices of S^3 and S^4 that make up the whitehead product $S^6 \to S^3 \vee S^4$)

This gives the attaching map of a sum of five whitehead products $S^6 \to S^3 \vee S^4$ with all distinct inclusion map. This is precisely the attaching map of $(S^3 \vee S^4)^{\#5}$ which shows that $\mathcal{Z}_\mathcal{K}$ is homotopic to $(S^3 \vee S^4)^{\#5}$.

7 Quasitoric Manifolds

7.1 The Davis-Januszkiewicz Classification Theorem

Definition 7.1.1 (Quasitoric Manifolds)
Let M be a smooth manifold of dimension $2n$. Let T^n be a torus acting smoothly on M. We say that M is a quasitoric manifold if the following are true.

- For each $p \in M$, there is a T^n-equivariant chart $(U, \varphi : U \to \mathbb{C}^n)$ where $\mathbb{C}^n$ is equipped with the standard torus action.
- There exists a convex polytope P such that there is a homeomorphism

$$M/T^n \cong P$$

Toric symplectic manifolds are quasitoric manifolds. Moment angle manifolds are NOT quasitoric manifolds.

Definition 7.1.2 (Orbit Type Strata)
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Let $H \leq T^n$ be a subgroup. Define the orbit type strata of H to be

$$M_{(H)} = \{x \in M \mid \text{Stab}_{T^n}(x) = H\}$$

Proposition 7.1.3
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Then there is a bijection

$$\{F \in \mathcal{F}(P) \mid F \text{ has codimension } k\} \overset{1:1}{\leftrightarrow} \left\{ N \subseteq M \;\middle|\; \begin{array}{c} N \text{ is an orbit type strata of subtori of} \\ T^n \text{ of dimension } k \end{array} \right\}$$

given as follows. Let $\pi : M \to P$ be the quotient map. Then we have

$$F_i \mapsto \pi^{-1}(\text{relint}(F))$$

Definition 7.1.4 (Primitive Vectors)
Let $v = (v_1, \dots, v_n) \in \mathbb{Z}^n$. We say that v is primitive if $v \neq \lambda u$ for any $\lambda \in \mathbb{Z} \setminus \{0\}$ and $u \in \mathbb{Z}^n$.

Definition 7.1.5 (The Characteristic Matrix)
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Define the characteristic matrix Λ as follows. For each facet F_i of P for $1 \leq i \leq m$, the facet corresponds to a a circle $S_i^1 \subseteq T^n$ by the above. Every circle $S_i^1 \subseteq T^n$ corresponds to a unique primitive vector $v_i = (v_{1i}, \dots, v_{1n}) \in \mathbb{Z}^n$ up to sign. Then

$$\Lambda = \begin{pmatrix} v_{11} & \cdots & v_{1m} \\ \vdots & \ddots & \vdots \\ v_{n1} & \cdots & v_{nm} \end{pmatrix} \in M_{n \times m}(\mathbb{Z})$$

Definition 7.1.6 (Characteristic Pairs)
Let P be a simple polytope of dimension n. Let $F_1, \dots, F_m$ be the facets of P. Let $\Lambda \in M_{n \times m}(\mathbb{Z})$ be a matrix. Let $c_1, \dots, c_m$ be the columns of Λ. We say that (P, Λ) is a characteristic pair if the following are true.

- c_i is a primitive vector for all $1 \leq i \leq m$.
- If $F_{i_1} \cap \cdots \cap F_{i_n}$ is a vertex of P, then

$$\det \begin{pmatrix} | & & | \\ c_{i_1} & \cdots & c_{i_n} \\ | & & | \end{pmatrix} = \pm 1$$

Definition 7.1.7 (The Associated Quasitoric Manifold)
Let (P, Λ) be a characteristic pair. Suppose that $\dim(P) = n$. Let $F_1, \dots, F_m$ be the facets of P. Let $c_1, \dots, c_m$ be the columns of Λ. Define the associated quasi-toric manifold of (P, Λ) by

$$M(P, \Lambda) = \frac{P \times T^n}{\sim}$$

where $(p, t_1) \sim (p, t_2)$ if $t_2^{-1}t_1 \in \langle c_i \mid p \in F(p) \rangle \cong T^k \leq T^n$ and $F(p)$ is the unique facet such that $p \in \text{relint}(F(p))$ for any $p \in P$.

Theorem 7.1.8 (Davis-Januszkiewicz Classification Theorem)
There is a one to one bijection

$$\frac{\{M \mid M \text{ is a quasitoric manifold}\}}{\cong} \overset{1:1}{\leftrightarrow} \frac{\{(P, \Lambda) \text{ is a characteristic pair}\}}{\sim}$$

where $(P, \Lambda) \sim (Q, \Gamma)$ if P and Q are combinatorially equivalent and there exists $A \in \text{GL}(n, \mathbb{Z})$ with $\det(A) = \pm 1$ such that $A\Lambda$ and Q are equal up to permutation of columns.

7.2 Basic Structures

Proposition 7.2.1
Let M be a quasitoric manifold with characteristic pair (P, Λ). Let $F_1, \dots, F_m$ be the facets of P. Then there is an isomorphism of graded rings

$$H^*(M; \mathbb{Z}) \cong \frac{\mathbb{Z}[x_1, \dots, x_m]}{I_{\text{SR}} + I_\Lambda}$$

where
- $\deg(x_i) = 2$ for $1 \leq i \leq m$.
- I_{SR} is the Stanley-Reisner Ideal given by

$$I_{\text{SR}} = (xj_1 \cdots x_{j_t} \mid F_{j_1} \cap \cdots \cap F_{j_t} = \emptyset)$$

- I_Λ is the linear relations given by

$$I_\Lambda = (\Lambda(x_1, \dots, x_n) = 0)$$

Recall that we denote $(h_0(P), \dots, h_n(p))$ by the h-vector of P.

Corollary 7.2.2
Let M be a quasitoric manifold with characteristic pair (P, Λ). Then we have

$$\text{rank}(H^{2k}(M; \mathbb{Z})) = h_k(P)$$

7.3 Bundles over Quasitoric Manifolds

Proposition 7.3.1
Let M be a quasitoric manifold. Then $TM \oplus \mathbb{R}^k$ admits a complex structure for some $k \in \mathbb{N}$.

The chern class of $TM \oplus \mathbb{R}^k$ is given by

$$c(TM \oplus \mathbb{R}^k) = \prod_{i=1}^{m}(1 + x_i) \in H^*(M) \cong \frac{\mathbb{Z}[x_1, \dots, x_m]}{I_{\text{SR}} + I_\Lambda}$$

There is a diffeo $Z_P/(G(P) \cong T^{m-n}) \cong M(P, \Lambda)$ for any P. (from this perspective one can see that projective toric manifolds are quasi-toric manifolds). Also this map $Z_P \to M(P, \Lambda)$ is a principle $G(P)$-bundle.

If $M(P, \Lambda) = V_P$ is a projective toric variety, the classical construction $\mathbb{C}^m \setminus Z(\Sigma_P)$ exhibits Z_P algebraically: there is an equivariant homotopy equivalence $\mathbb{C}^m \setminus Z(\Sigma_P) \simeq Z_P$.

8 Kahler Manifolds

8.1 Equivalent Definitions

Definition 8.1.1 (Kahler Forms)
Let (M, h) be a complex Hermitian manifold. Let ω be the associated form. We say that ω is a Kahler form if $d\omega = 0$. In this case, we say that M is a Kahler manifold.

Lemma 8.1.2
Let M be a Kahler manifold. Then the following are true.
- Let $g = \frac{1}{2}(h + \overline{h})$ be the associated Riemannian metric. Then (M, g) is a Riemannian manifold.
- Let $\omega = \frac{i}{2}(h - \overline{h})$ be the associated $(1, 1)$-form. Then (M, ω) is a symplectic manifold.

Proposition 8.1.3
Let (M, ω) be a symplectic manifold. Then ω is a Kahler form if and only if M admits an integrable almost complex structure J. In this case, the following are true.
- The Riemannian metric is given by $g(u, v) = \omega(u, Jv)$.
- The Hermitian metric is given by $h = g - i\omega$.

Proposition 8.1.4
Let (M, g) be a Riemannian manifold of dimension $2n$. Then M admits a Kahler form if and only if $\text{Hol}(\nabla^g) \leq U(n)$.

Example 8.1.5
$\mathbb{CP}^n$ together with the Fubini-Study metric is a Kahler manifold.

Example 8.1.6
Every Riemann surfaces is a Kahler manifold.

8.2 Basic Structures

Theorem 8.2.1 (Banyaga)
Let M be a compact complex manifold. Let ω_1 and ω_2 be Kahler forms on M. Then (M, ω_1) and (M, ω_2) are symplectomorphic.

8.3 Main Theorems

Theorem 8.3.1 (Hodge Decomposition Theorem)
Let M be a compact Kahler manifold. Then the following are true.
- There is an isomorphism
$$H^k_{\text{dR}}(M; \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(M)$$
for all $k \in \mathbb{N}$.
- There is an isomorphism
$$H^{p,q}(M) \cong \overline{H^{q,p}(M)}$$
for all $p, q \in \mathbb{N}$.
- $\dim_{\mathbb{R}}(H^k_{\text{dR}}(M))$ is even for all $k \in \mathbb{N}$.

Example 8.3.2
Every compact Riemann surface is of the form Σ_g for some $g \geq 0$. We have seen above that Σ_g is

Kahler. We know that

$$H^k_{\mathrm{dR}}(\Sigma_g;\mathbb{C}) = \begin{cases} \mathbb{C} & \text{if } k = 0, 2 \\ \mathbb{C}^{2g} & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

From this we conclude that $H^{0,0}(\Sigma_g) = \mathbb{C}$. Now since $\dim_{\mathbb{C}}(H^{p,q}(M)) = \dim_{\mathbb{C}}(\overline{H^{q,p}(\Sigma_g)}) = \dim_{\mathbb{C}}(H^{q,p}(\Sigma_g))$, we conclude that $H^{1,0}(\Sigma_g) \cong H^{0,1}(\Sigma_g) \cong \mathbb{C}^g$. Also, since Σ_g has complex dimension 1, it has no holomorphic 2-forms. Hence $H^{2,0}(\Sigma_g) = H^{0,2}(\Sigma_g) = 0$. This leaves us with $H^{1,1}(\Sigma_g) = H^2(\Sigma_g) = \mathbb{C}$. The hodge diamond of Σ_g is given by

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Definition 8.3.3 (Hodge Filtration)
Let M be a compact Kahler manifold. Define the Hodge filtration of M to be

$$F^i H^k_{\mathrm{dR}}(M;\mathbb{C}) = \bigoplus_{p \geq i} H^{p,q}(M)$$

Definition 8.3.4 (The Lefschetz Operator)
Let M be a Kahler manifold. Let ω be its associated form. Define the Lefschetz operator to be the map $L : \Omega^k(M) \to \Omega^{k+2}(M)$ given by

$$\alpha \mapsto \alpha \wedge \omega$$

Theorem 8.3.5 (Hard Lefschetz Theorem)
Let M be a compact Kahler manifold. Then the following are true.

- The Lefschetz operator induces an isomorphism

$$L^k : H^{n-k}_{\mathrm{dR}}(M) \xrightarrow{\cong} H^{n+k}_{\mathrm{dR}}(M)$$

Definition 8.3.6 (Primitive Cohomology)
Let M be a compact Kahler manifold. Let $n = \dim(M)$. Define the primitive cohomology of M to be

$$H^i_{\text{prim}}(M) = \ker(L^{n-i+1} : H^i_{\mathrm{dR}}(M) \to H^{2n-i+2}_{\mathrm{dR}}(M))$$

Definition 8.3.7 (Lefschetz Decomposition)
Let M be a compact Kahler manifold. Define the Lefschetz decomposition of M to be

$$H^k_{\mathrm{dR}}(M) = \bigoplus_{2r \leq k} L^r(H^{k-2r}_{\text{prim}}(M))$$

Proposition 8.3.8
Let M be a compact Kahler manifold of complex dimension $2n$. Suppose that $h = \sum_{p=0}^{2m}\sum_{q=0}^{2m}(-1)^p \dim_{\mathbb{C}}(H^{p,q}(M))$. Then the signature of the non-degenerate symmetric bilinear form $H^n(X;\mathbb{R}) \times H^n(X;\mathbb{R}) \to \mathbb{R}$ from Poincare duality is given by $(h, 2n-h)$.

8.4 Line Bundles on Kahler Manifolds

Theorem 8.4.1 (Lefschetz Theorem on $(1,1)$-Classes)
Let M be a compact Kahler manifold. Then we have

$$\text{im}(\text{Pic}(M) \to H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{C})) = \text{im}(H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{C})) \cap H^{1,1}(M)$$

9 Algebra

9.1 Algebra, Coalgebra and Bialgebra

Let R be a commutative ring. Recall that an R-algebra is a ring A together with a ring homomorphism $R \to A$. Equivalently, it is an R-module with multiplication that is associative and respects the R-module structure.

Proposition 9.1.1
Let R be a commutative ring. Let $A \in \mathbf{Mod}_R$ be an R-module. Let $m : A \otimes_R A \to A$ and $u : R \to A$ be R-module homomorphisms. Then (A, m, u) is an associative algebra with multiplication given by m and identity given $u(1)$ if and only if the following diagrams commute:

$$\begin{tikzcd}
(A \otimes_R A) \otimes_R A \arrow[r, "m \otimes 1"] \arrow[d, "\alpha"'] & A \otimes_R A \arrow[r, "m"] & A \\
A \otimes_R (A \otimes_R A) \arrow[r, "1 \otimes m"'] & A \otimes_R A \arrow[ur, "m"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
R \otimes_R A \arrow[r, "u \otimes \mathrm{id}_A"] \arrow[dr, "\lambda"'] & A \otimes_R A \arrow[d, "m"'] & A \otimes R \arrow[l, "\mathrm{id}_A \otimes u"'] \arrow[dl, "\mu"] \\
& A &
\end{tikzcd}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Equivalently, the above proposition is the same as saying (A, m, u) is a monoid object in the monoidal category $(\mathbf{Mod}_R, \otimes_R, R)$. Motivated by the categorical description of an R-algebra, we define coalgebras as the formal dual of the above object in the following way.

Definition 9.1.2 (Coalgebras)
Let R be a commutative ring. Let $C \in \mathbf{Mod}_R$ be an R-module. Let $\Delta : C \to C \otimes_R C$ and $\varepsilon : C \to R$ be R-module homomorphisms. We say that (C, Δ, ε) is a coalgebra if the following diagrams commute:

$$\begin{tikzcd}
C \arrow[r, "\Delta"] \arrow[dr, "\Delta"'] & C \otimes_R C \arrow[r, "\Delta \otimes 1"] & (C \otimes_R C) \otimes_R C \arrow[d, "\alpha"] \\
& C \otimes_R C \arrow[r, "1 \otimes \Delta"'] & C \otimes_R (C \otimes_R C)
\end{tikzcd}
\qquad
\begin{tikzcd}
& C \arrow[dl, "\lambda"'] \arrow[d, "\Delta"] \arrow[dr, "\mu"] & \\
R \otimes_R C & C \otimes C \arrow[l, "\varepsilon \otimes \mathrm{id}_A"] \arrow[r, "\mathrm{id}_A \otimes \varepsilon"'] & C \otimes_R R
\end{tikzcd}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Definition 9.1.3 (Bialgebras)
Let R be a commutative ring. Let $B \in \mathbf{Mod}_R$ be an R-module. Let $m : B \otimes_R B \to B$, $u : R \to B$, $\Delta : B \to B \otimes_R B$ and $\varepsilon : B \to R$ be R-module homomorphisms. We say that B is an R-bialgebra if the following are true.

- (B, m, u) is an R-algebra.
- (B, Δ, ε) is an R-coalgebra.
- Multiplication and comultiplication are compatible:

$$\begin{tikzcd}
B \otimes_R B \arrow[r, "m"] \arrow[d, "\Delta \otimes \Delta"'] & B \arrow[r, "\Delta"] & B \otimes_R B \\
B \otimes_R B \otimes_R B \otimes_R B \arrow[rr, "\mathrm{id} \otimes \tau \otimes \mathrm{id}"'] & & B \otimes_R B \otimes_R B \otimes_R B \arrow[u, "m \otimes m"']
\end{tikzcd}$$

  where τ is the natural isomorphism from commutativity of tensor products.
- Multiplication and counit are compatible:

$$\begin{tikzcd}
B \otimes_R B \arrow[r, "m"] \arrow[d, "\varepsilon \otimes \varepsilon"'] & B \arrow[d, "\varepsilon"] \\
R \otimes_R R \arrow[r, "\lambda"'] & R
\end{tikzcd}$$

  where λ is the natural isomorphism from unitality of tensor products.

- Comultiplication and unit are compatible:

$$\begin{tikzcd} R \arrow[r, "u"] \arrow[d, "\lambda"'] & B \arrow[d, "\Delta"] \\ R \otimes_R R \arrow[r, "\Delta\otimes\Delta"'] & B \otimes_R B \end{tikzcd}$$

where λ is the natural isomorphism from unitality of tensor products.
- Unit and counit are compatible:

$$\begin{tikzcd} R \arrow[rr, "\mathrm{id}"] \arrow[dr, "u"'] & & R \\ & B \arrow[ur, "\varepsilon"'] & \end{tikzcd}$$

9.2 Hopf Algebra

Definition 9.2.1 (Hopf Algebras)
Let R be a commutative ring. Let $(H, m, u, \Delta, \varepsilon)$ be an R-bialgebra. Let $S : H \to H$ be an R-module homomorphism. We say that $(H, m, u, \Delta, \varepsilon, S)$ is a Hopf algebra if the following diagram commutes:

$$\begin{tikzcd} & H \otimes_R H \arrow[rr, "S\otimes\mathrm{id}_H"] & & H \otimes_R H \arrow[dr, "m"] & \\ H \arrow[ur, "\Delta"] \arrow[rr, "\varepsilon"] \arrow[dr, "\Delta"'] & & R \arrow[rr, "u"] & & H \\ & H \otimes_R H \arrow[rr, "\mathrm{id}_H\otimes S"'] & & H \otimes_R H \arrow[ur, "m"'] & \end{tikzcd}$$

Definition 9.2.2 (Commmutative and Cocommutative)
Let R be a commutative ring. Let H be a Hopf algebra.
- We say that H is commutative if the underlying algebra is commutative.
- We say that H is cocommutative if the underlying coalgebra is cocommutative.

Proposition 9.2.3
Let R be a commutative ring. Let H be a Hopf algebra. If H is a projective and finitely generated R-modue, then $\mathrm{Hom}_R(H, R)$ is a Hopf algebra. Moreover, the following are true.
- $\mathrm{Hom}_R(M, R)$ is commutative if and only if H is cocommutative.
- $\mathrm{Hom}_R(M, R)$ is cocommutative if and only if H is commutative.

9.3 Differential Graded Algebras

Definition 9.3.1 (Differential Graded Algebras)
Let R be a commutative ring. A differential graded R-algebra consists of the following data.
- A $\mathbb{Z}$-graded R-algebra A.
- An R-algebra homomorphism of degree -1 (called homological dgas) or of degree 1 (called cohomological dgas).

such that the following are true.
- Differential: $d \circ d = 0$.
- Graded Leibniz: $d(ab) = (da)b + (-1)^{\deg(a)}a(db)$ for a, b homogeneous elements of A.

The differential property means that any (homological) differential graded algebra decomposes into a chain complex whose differential is given by d.

Definition 9.3.2 (Homomorphisms of Differential Graded Algebras)
Let R be a commutative ring. Let A, B be differential graded algebras. A morphism $\varphi : A \to B$ of differential graded algebras is a graded R-algebra homomorphism.

Lemma 9.3.3
Let R be a commutative ring. Let (A, d) be a cohomological differential graded algebra. Then $H^*(A, d) = \bigoplus_{n\in\mathbb{Z}} H^n(A, d)$ is a graded ring with multiplication defined by

$$[u] \cdot [v] = [uv]$$

for $u \in H^n(A, d)$ and $v \in H^m(A, d)$.

Proof
Notice that by the graded Leibniz rule, we have $d(uv) = (du)v + (-1)^n ud(v) = 0$ since u and v are cocycles. Hence uv is also a cocycle. ■

Products of DGA, tensor products, hom set

Definition 9.3.4 (Commutative Differential Graded Algebras)
Let R be a commutative ring. Let A be a differential graded algebra. We say that A is commutative (cdga) if A is graded commutative.

Lemma 9.3.5
Let R be a commutative ring. Let M be an R-module. Let $\varphi : M \to R$ be an R-module homomr-phism. Then $K_\bullet(\varphi)$ is a commutative differential graded algebra.

Proposition 9.3.6
Let k be a field. Let A, B be cohomological differential graded algebras. Then the following are true.

- There is a natural isomorphism $H^*(A) \otimes_k H^*(B) \cong H^*(M \otimes_k N)$.
- There is a natural isomorphism $H^*(\text{Hom}_k(A, B)) \cong \text{Hom}_k(H^*(A), H^*(B))$

9.4 Minimal DGAs

Definition 9.4.1 (Free Graded-Commutative Algebra)
Let R be a commutative ring. Let M be a graded R-module. Define the free graded-commutative algebra on M by

$$E(M) = \text{Sym}\left(\bigoplus_{i\in 2\mathbb{Z}} M_i\right) \oplus \Lambda\left(\bigoplus_{i\in 2\mathbb{Z}+1} M_i\right)$$

Definition 9.4.2
Let R be a ring. Let M be a graded R-module. A minimal DGA over M is a differential graded algebra $(E(M), d)$ such that $d(M) \subseteq E(M)_2$.

9.5 Massey Products

Definition 9.5.1 (Massey Triple Products)
Let R be a commutative ring. Let A be a differential graded algebra. Let $u \in A^p$, $v \in A^q$ and $w \in A^r$. Suppose that $[u] \cdot [v] = [v] \cdot [w] = 0$. Define the Massey triple product of the three elements to be

$$\langle [u], [v], [w] \rangle = \{[\overline{s} \cdot w + \overline{u} \cdot t] \in H^{p+q+r-1}(A) \mid s \in A^{p+q-1}, t \in A^{q+r-1} \text{ and } ds = \overline{u} \cdot v, dt = \overline{v} \cdot w\}$$

where $\overline{a} = (-1)^{1+\deg(a)} a$ for any homogeneous element $a \in A$.

Lemma 9.5.2
Let R be a commutative ring. Let A be a differential graded algebra. Let $u \in A^p$, $v \in A^q$ and $w \in A^r$. Suppose that $[u] \cdot [v] = [v] \cdot [w] = 0$. Then we have

$$\langle [u], [v], [w] \rangle = \frac{H^{p+q+r-1}(A)}{[u] \cdot H^{q+r-1}(A) + H^{p+q-1}(A) \cdot [w]}$$

Proposition 9.5.3
Let $p, q, r \in \mathbb{N}$. Let $Y = S^p \vee S^q \vee S^r$. Let $\alpha \in \pi_p(Y)$, $\beta \in \pi_q(Y)$ and $\gamma \in \pi_r(Y)$ be generators. Let $X = Y \cup_{[\alpha,[\beta,\gamma]]} D^{p+q+r-1}$. Let $u, v, w, z \in H^*(X)$ be generators of degree $p, q, r, p+q+r-1$ respectively. Then the following are true.

- $\langle u, v, w \rangle = (-1)^p z$.
- $\langle v, w, u \rangle = (-1)^{pq+pr+p} z$.
- $\langle w, u, v \rangle = 0$.

10 Koszul Algebras and Quadratic Algebras

11 Profinite Groups

11.1 Basic Properties

Definition 11.1.1 (Profinite Groups)
Let G be a topological group. We say that G is a profinite group if there exists an inverse system $X : \mathcal{I} \to {}_G\textbf{Top}$ consisting of finite groups with the discrete topology such that

$$G = \lim_{\mathcal{I}} X$$

In particular, this means that G is equipped with the weak topology.

Proposition 11.1.2
Let G be a topological group. Then G is profinite if and only if G is compact and totally disconnected.

Corollary 11.1.3
Let G be a profinite group. Then the following are true.
- G is Haudorff.
- G is locally compact.

Proposition 11.1.4
Let G be a compact topological group. Then G/G_0 is a profinite group.

Proposition 11.1.5
Let G be a profinite group. Let $H \leq G$ be a subgroup. Then the following are true.
- If H is closed, then H is profinite.
- If H is closed and normal, then G/H is profinite.
- H is open if and only if H is closed and $[G : H] < \infty$.

11.2 Profinite Completion

12 Equivariant Algebraic Topology

12.1 G-Homotopy

Definition 12.1.1 (G-Homotopy) Let G be a topological group and let X, Y be G-spaces. Let $f, g : X \to Y$ be G-equivariant maps. A G-homotopy from f to g is a homotopy $H : X \times I \to Y$ from f to g such that for each $t \in I$, the map

$$H(-, t) : X \to Y$$

is G-equivariant map.

12.2 Equivariant CW Complexes

12.3

Definition 12.3.1 (The Homotopy Quotient (Borel Construction))
Let G be a topological group. Let X be a G-space. Define the homotopy quotient of X to be the orbit space

$$EG \times_G X = \frac{EG \times X}{G}$$

where the action of G on $EG \times X$ is given by $g \cdot (e, x) = (g \cdot e, g^{-1} \cdot x)$.

Proposition 12.3.2
Let G be a topological group. Let X be a G-space. Then

$$X \hookrightarrow EG \times_G X \to BG$$

is a fibration.

Lemma 12.3.3
Let G be a topological group. Let X be a G-space. If G acts freely on X, then there is a homotopy equivalence

$$EG \times_G X \simeq \frac{X}{G}$$

induced by the projection map $EG \times_G X \to X$.

13

There is a one to one bijection between representations of $\pi_1(X)$ and locally constant sheaves (called local systems).

Definition 13.0.1
Let X be a space. Let $x_0 \in X$. Define the functor $\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\textbf{Sets}$ as follows.

- For $p : \widetilde{X} \to X$ a covering space, define $\text{Fib}_{x_0}(p) = p^{-1}(x_0)$.
- For $f : \widetilde{X}_1 \to \widetilde{X}_2$ a map of covering spaces, define $\text{Fib}_{x_0}(f)$ to be $f|_{p^{-1}(x_0)}$.

Theorem 13.0.2
Let X be a connected and locally simply connected space. Let $x_0 \in X$. Then

$$\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\textbf{Sets}$$

is fully faithful and essentially surjective.

Theorem 13.0.3 (Classification of Local Systems)
Let X be a space (locally compact, Hausdorff, second countable, locally path connected, semi locally path connected, path connected). Then there is an equivalent of categories

$$\text{Func}(\textbf{Open}^{\text{op}}, \text{Vect}_{\mathbb{C}}) \simeq \text{Rep}(\pi_1(X))$$

Ref: Galois Groups and Fundamental Groups

Theorem 13.0.4 (Classification Theorem for Flat Connections)
Let M be a connected Riemannian manifold. Let G be a Lie group. Then there is a one to one correspondence

$$\frac{\{(P, A) \mid P \to M \text{ is a principal } G\text{-bundle and } A \text{ is a flat connection}\}}{\sim} \overset{1:1}{\longleftrightarrow} \text{Rep}(\pi_1(X), G)$$

where $(P_1, A_1) \sim (P_2, A_2)$ if there exists bundle isomorphism $\varphi : P_1 \to P_2$ such that $\varphi^* A_2 = A_1$.

Ref: Differential Geometry; Taubes Bijection described in: Geometry of Characteristic classes; Morita

There is a natural bijection between principal bundles and vector bundles by taking the associated bundle. If $G = GL(V)$ for some finite dimensional vector space V over the field $k = \mathbb{R}$ or $\mathbb{C}$, the sheaf of sections of the associated bundle is a locally constant sheaf.

Also: Upgrade to diffeomorphism.

14 Infinite Galois Theory

Proposition 14.0.1
Let F be a field. Let K/F be a Galois extension. Then the following are true.

- The inverse limit of the system $\text{Gal}(L/F)$ for L/F a finite Galois extension is $\text{Gal}(K/F)$.
- $\text{Gal}(K/F)$ is a profinite group.
- The projection maps $\text{Gal}(K/F) \to \text{Gal}(L/F)$ for L/F a finite Galois extension is a surjective homomorphism.
- Let $F < L < K$ be a field extension. Then K/L is a Galois extension and $\text{Gal}(K/L)$ is a closed subgroup of $\text{Gal}(K/F)$.

Theorem 14.0.2 (Krull)

Let F be a field. Let K/F be a field extension. Then there is an inclusion reversing bijection

$$\begin{Bmatrix} \text{Closed subgroups of} \\ \text{Aut}_F(K) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Intermediary subfields} \\ F<L<K \end{Bmatrix}$$

given as follows.

- For a subgroup $H \leq \text{Aut}_F(K)$, $H \mapsto \text{Fix}_F(H)$.
- For an intermediary subfield M, $M \mapsto \text{Aut}_L(K)$.

Proposition 14.0.3

Let F be a field. Let K/F be a Galois extension. Then the above bijection restricts to a bijection

$$\begin{Bmatrix} \text{Normal Subgroups} \\ \text{of Aut}_K(L) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Normal Extensions} \\ K<M \end{Bmatrix}$$

15 Local Cohomology

15.1

Definition 15.1.1 (The I-Torsion of a Module)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Define the I-torsion of M to be

$$\Gamma_I(M) = \{m \in M \mid I^t m = 0 \text{ for some } t \in \mathbb{N}\}$$

Definition 15.1.2 (I-Torsion Modules)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. We say that M is I-torsion if $M = \Gamma_I(M)$.

Definition 15.1.3 (The Torsion Functor)
Let R be a Noetherian commutative ring. Let I be an ideal of R. Define the I-torsion functor

$$\Gamma_I : {}_R\mathbf{Mod} \to {}_R\mathbf{Mod}$$

by the following.
- For each R-module M, the functor sends M to the I-torsion $\Gamma_I(M)$.
- For each R-module homomorphism $f : M \to N$, the functor sends f to $\Gamma_I(f) = f|_{\Gamma_I(M)}$.

Lemma 15.1.4
Let R be a Noetherian commutative ring. Let I be an ideal of R. Then Γ_I is a left exact functor.

Definition 15.1.5 (Local Cohomology)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Define the nth local cohomology of M with support in I to be

$$H_I^n(M) = (R^n\Gamma_I)(M)$$

Proposition 15.1.6
Let R be a Noetherian commutative ring. Let M be an R-module. Let I, J be ideals of R. If $\sqrt{I} = \sqrt{J}$, then the equality $\Gamma_I(M) = \Gamma_J(M)$ induces an equality

$$H_I^n(M) = H_J^n(M)$$

for all $n \in \mathbb{N}$.

Thus we only need to consider the local cohomology with support in radical ideals.

Lemma 15.1.7
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Then $H_I^n(M)$ is I-torsion for all $n \in \mathbb{N}$.

Lemma 15.1.8
Let R be a Noetherian commutative ring. Let M_i be R-modules for each $i \in J$. Let I be an ideal of R. Then there is a natural isomorphism

$$H_I^n\left(\bigoplus_{i\in J} M_i\right) \cong \bigoplus_{i \in J} H_J^n(M_i)$$

Proposition 15.1.9
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R.Then there are natural isomorphisms

$$\varinjlim_{i\in\mathbb{N}} \mathrm{Ext}^n_R(R/I^i, M) \cong H^n_I(M)$$

for all $n \in \mathbb{N}$.

16 Morita Theory

16.1

Definition 16.1.1 (Morita Equivalence)
Let R, S be rings. We say that R and S are Morita equivalent if there exists an equivalence of categories
$$_R\mathbf{Mod} \simeq {_S\mathbf{Mod}}$$

Lemma 16.1.2
Let R, S be rings. Then R and S are Morita equivalent if and only if there exists an equivalence of categories
$$\mathbf{Mod}_R \simeq \mathbf{Mod}_S$$

Lemma 16.1.3
Let R, S be rings. Suppose that R and S are Morita equivalent. Let $F : {_R\mathbf{Mod}} \to {_S\mathbf{Mod}}$ be an equivalence. Then F is additive.

16.2 Progenerators

Definition 16.2.1 (Progenerators)
Let R be a ring. Let P be an R-module. We say that P is a progenerator of R if the following are true.
- P is finitely generated.
- P is projective.

Proposition 16.2.2
Let R, S be rings. Let $F : {_R\mathbf{Mod}} \to {_S\mathbf{Mod}}$ and $G : {_S\mathbf{Mod}} \to {_R\mathbf{Mod}}$ be additive functors. Then F and G are inverses if and only if there exists an (S, R)-bimodule P such that the following are true.
- $_SP$ is a progenerator.
- P_R is a progenerator.
- There are natural isomorphisms $F \cong P \otimes_R -$ and $G \cong \text{Hom}(_SP, -)$.

Theorem 16.2.3 (Morita)
Let R, S be rings. Then the following are equivalent.
- R and S are Morita equivalent.
- There exists a progenerator R-module P_R such that $S \cong \text{End}_R(P_R)$.
- There exist an (R, S)-bimodule M and a (S, R)-bimodule N such that $M \otimes_S N \cong R$ as (R, R)-bimodules and $N \otimes_R M \cong S$ as (S, S)-bimodules.

16.3 Properties of Morita Equivalence

Proposition 16.3.1
Let R, S be rings. If R and S are Morita equivalent, then the following are true.
- There is a ring isomorphism $Z(R) \cong Z(S)$.
- $R/J(R)$ is Morita equivalent to $S/J(S)$.

Proposition 16.3.2
Let R, S be rings. Let M be an R-module. Suppose that R and S are Morita equivalent via the functor $F : {_R\mathbf{Mod}} \to {_S\mathbf{Mod}}$. Then the following are true.
- M is injective if and only if $F(M)$ is injective.
- M is projective if and only if $F(M)$ is projective.
- M is simple if and only $F(M)$ is simple.

- M is semisimple if and only if $F(M)$ is semisimple.
- M is finitely generated if and only if $F(M)$ is finitely generated.
- M is Noetherian if and only if $F(M)$ is Noetherian.
- M is Artinian if and only if $F(M)$ is Artinian.

Proposition 16.3.3
Let R, S be rings. Suppose that R and S are Morita equivalent. Then the following are equivalent.
- R is simple if and only if S is simple.
- R is semisimple if and only if S is semisimple.
- R is Noetherian if and only if S is Noetherian.
- R is Artinian if and only if S is Artinian.

Proposition 16.3.4
Let R, S be rings. Then R and S are Morita equivalent if and only if ${}_R\mathbf{FGMod} \simeq {}_S\mathbf{FGMod}$.

17 Complex Analytic Space

17.1 Complex Analytic Varieties

Definition 17.1.1 (Complex Analytic Varieties)
Let $Z \subseteq \mathbb{C}^n$. We say that Z is a complex analytic variety if there exists an open set $Z \subseteq U \subseteq \mathbb{C}^n$ and $f_1, \dots, f_k \in \mathcal{O}_{\mathbb{C}^n}(U)$ such that
$$Z = \mathbb{V}(f_1, \dots, f_k)$$

We use a subset of C^n in the definition because in $\mathbb{C}^n$, locally defined functions cannot in general be defined globally (compare it to the real case, where they have partitions of unity and bump functions).

Indeed, for smooth functions on $\mathbb{R}^n$, a function on an open set U can be modified (via bump functions) to produce a global function that agrees with it on a smaller open subset. You can always find a global function that captures the local behavior near any point.

For holomorphic functions on $\mathbb{C}^n$: no such modification is possible. A holomorphic function cannot be cut off or modified while remaining holomorphic, so local holomorphic data is genuinely trapped in its domain.

Definition 17.1.2 (The Structure Sheaf) Let$U \subseteq \mathbb{C}^n$ be an open subset. Let $f_1, \dots, f_k \in \mathcal{A}_{\mathbb{C}^n}(U)$ be analytic functions on U. Let $Z = \mathbb{V}(f_1, \dots, f_k)$ be the vanishing locus of $f_1, \dots, f_k$. Define the structure sheaf
$$\mathcal{O}_Z : \textbf{Open}(Z) \to \textbf{CRing}$$
as follows.

- For each open set $U \subseteq Z$, define
$$\mathcal{O}_Z(U) = \frac{\mathcal{O}_{\mathbb{C}^n}(U)}{(f_1, \dots, f_k)}$$
- For each inclusion $U \subseteq V \subseteq Z$ of open sets, define
$$\mathcal{O}_Z(V) \to \mathcal{O}_Z(U)$$
to be the induced map of the restriction $\mathcal{O}_{\mathbb{C}^n}(V) \to \mathcal{O}_{\mathbb{C}^n}(U)$.

17.2 Complex Analytic Subvarieties of Manifolds

Definition 17.2.1 (Complex Analytic Subvarieties)
Let M be a complex manifold. Let $Z \subseteq M$ be a subset. We say that Z is a complex analytic subvariety of M if for each $p \in Z$, there exists a chart (U, ϕ) of M such that $\phi(Z \cap U)$ is a complex analytic variety.

In particular, complex analytic subvarieties of $\mathbb{C}^n$ are just complex analytic varieties.

Theorem 17.2.2 (Cartan's (Coherence) Theorem) Let M be a complex manifold and let A be an analytic subset of M. Then $\mathcal{I}_A$ is a coherent sheaf of $\mathcal{O}_M$-modules.

17.3 Complex Analytic Spaces

Definition 17.3.1 (Complex Analytic Spaces) Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is a complex analytic space if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is a complex analytic space.

Lemma 17.3.2
Let $(X, \mathcal{O}_X)$ be a complex analytic space. Then X is a complex manifold if and only if for each $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is an isomorphism

$$(U, \mathcal{O}_X|_U) \cong (V, \mathcal{O}_{\mathbb{C}^n}|_V)$$

for some open subset $V \subseteq \mathbb{C}^n$.

Theorem 17.3.3 Let X be a complex space. Then X_{reg} is dense and open in X. Moreover, X_{reg} consists of a disjoin union of complex manifolds.

Theorem 17.3.4 Let X be an irreducible complex space. Then every non-constant holomorphic function $f : X \to \mathbb{C}$ is an open map.

Corollary 17.3.5 Let X be an irreducible compact complex space. Then every holomorphic function $f : X \to \mathbb{C}$ on X is constant.

Definition 17.3.6 (Complex Analytic Subvarieties)
Let $(X, \mathcal{O}_X)$ be a complex analytic space. Let $Z \subseteq X$ be subset. We say that Z is a complex analytic subvariety if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ and $f_1, \dots, f_k \in \mathcal{O}_X(U)$ such that

$$Z \cap U = \mathbb{V}_U(f_1, \dots, f_k)$$

This definition generalizes complex analytic subvarieties of complex manifolds.

17.4 Connections to Algebraic Geometry

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

Definition 17.4.1 (The Analytification of a Scheme)
Let X be a scheme locally of finite type over $\mathbb{C}$. Define the analytification of X to be the unique analytic space X^{an} together with a morphism of locally ringed space $\alpha : X^{\text{an}} \to X$ satisfying the following universal property: For any analytic space Y and any morphism of locally ringed spaces $f : Y \to X$, there exists a unique morphism of analytic spaces $g : Y \to X^{\text{an}}$ such that the following diagram commutes:

$$\begin{tikzcd} & Y \arrow[dl, dashed, "\exists! g"'] \arrow[d, "f"] \\ X^{\text{an}} \arrow[r, "\alpha"'] & X \end{tikzcd}$$

Recall that the Yoneda embedding $h : \textbf{LRS} \to \textbf{Set}^{\textbf{LRS}}$ is the functor sending a locally ringed space X to the functor $h_X : \textbf{LRS} \to \textbf{Set}$ defined by $h_X(Y) = \text{Hom}_{\textbf{LRS}}(Y, X)$. Then the universal property of the analytification means that the functor h_X restricted to analytic spaces $\textbf{An}$ is representable.

One thinks of the morphism $X^{\text{an}} \to X$ as doing two things. It is an enlargement of the standard topology of X^{an} to the Zariski topology of X, and it introduces generic points outside of $X(\mathbb{C})$ (Indeed we will see that X^{an} has the same underlying set as $X(\mathbb{C})$).

Theorem 17.4.2
Let X be a scheme locally of finite type over $\mathbb{C}$. Then the analytification of X exists and is unique up to unique isomorphism.

Definition 17.4.3 (The Analytification Functor)
Define the analytification functor
$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \to \mathbf{An}$$
as follows.
- For each scheme X locally of finite type over $\mathbb{C}$, we have X^{an} is the analytification of X.
- For a morphism of schemes $f : X \to Y$ over $\mathbb{C}$, the universal propertyof Y^{an} induces a morphism $X \to Y^{\text{an}}$. Precomposing with the canonical morphism $X^{\text{an}} \to Y^{\text{an}}$ gives the desired map $X^{\text{an}} \to Y^{\text{an}}$.

Proposition 17.4.4
Let X be a scheme locally of finite type over $\mathbb{C}$. Then the underlying set of X^{an} is given by $X(\mathbb{C})$.

Proposition 17.4.5
Let X be a scheme locally of finite type over $\mathbb{C}$. Then the following are true.
- X is separated if and only if X^{an} is Hausdorff.
- X is connected if and only if X^{an} is connected.
- X is smooth if and only if X^{an} is a complex manifold.
- X is proper if and only if X^{an} is compact.

Proposition 17.4.6
Let X be a variety over $\mathbb{C}$. Then the following are true.
- If X is irreducible, then X^{an} is connected.
- If X is smooth and affine, then X^{an} is a Stein manifold.
- If X is smooth and projective, then X^{an} is Kahler.

Theorem 17.4.7 (GAGA)
Let X be a proper scheme of finite type over $\mathbb{C}$. Then the following are true.
- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then $\mathcal{F}^{\text{an}}$ is a sheaf of $\mathcal{O}_{X^{\text{an}}}$-module.
- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. There is a natural isomorphism
$$H^i(X, \mathcal{F}) \cong H^i(X^{\text{an}}, \mathcal{F}^{\text{an}})$$
- The analytification functor induces an equivalence of categories
$$\mathbf{Coh}_{\mathcal{O}_X} \cong \mathbf{Coh}_{\mathcal{O}_{X^{\text{an}}}}$$
- Let $Y \subseteq X^{\text{an}}$ be a closed complex analytic subspace. Then there exists a closed subvariety Z of X such that $Z^{\text{an}} = Y$.

Corollary 17.4.8 (Chow's Theorem)
Let X be a closed complex analytic subspace of $\mathbb{CP}^n$. Then X is a projective variety over $\mathbb{C}$.

The important point is that: the theory of smooth projective complex varieties are subsumed by the theory of Kahler manifolds. Divisors and line bundles correspond bijectively thanks to Chow's theorem.

Theorem 17.4.9 (Grothendieck)
Let X be a smooth variety over $\mathbb{C}$. Then there is a natural isomorphism
$$H^n_{\text{dR}}(X) \cong H^n_{\text{Holo}}(X^{\text{an}}) \cong H^n_{\text{dR}}(X^{\text{an}})$$
induced by the canonical map $\alpha : X^{\text{an}} \to X$.

Corollary 17.4.10
Let X be a smooth variety over $\mathbb{C}$. Then the above isomorphism and de Rham's theorem induces

a natural isomrophism

$$H^n_{\mathrm{dR}}(X) \cong H^n_{\mathrm{sing}}(X^{\mathrm{an}}; \mathbb{C})$$

Theorem 17.4.11 (The Period Isomorphism)
Let k be a subfield of $\mathbb{C}$. Let X be a smooth variety over k. Then there is a natural isomorphism

$$H^*_{\mathrm{dR}}(X) \otimes_k \mathbb{C} \cong H^*_{\mathrm{sing}}(X^{\mathrm{an}}; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

18 Divisors and Line Bundles

18.1 Divisors

Definition 18.1.1 (Divisors on a Complex Manifold)
Let X be a complex manifold. A divisor on X is a formal linear combination

$$D = \sum_{\substack{V \text{ is an analytic variety of } X \\ \text{of dimension } \dim(X)-1}} n_V \cdot V$$

where $n_V \in \mathbb{Z}$, and such that for each $p \in X$, there exists an open neighbourhood U of p such that U meets finitely many V in the summand.

The locally finite condition here is required because in Algebraic Geometry, Noetherianess is assumed, otherwise they are the same.

Definition 18.1.2 (Group of Divisors)
Let X be a complex manifold. Define the group of divisors of X to be the set

$$\text{Div}(X) = \{D \mid D \text{ is a divisor on } X\}$$

together with addition given as a subgroup of $\prod_V \mathbb{Z} \cdot V$.

Definition 18.1.3 (Effective Divisors)
Let X be a complex manifold. Let $D = \sum_V n_V \cdot V \in \text{Div}(X)$ be a divisor. We say that D is effective if $n_V \geq 0$ for all V.

Definition 18.1.4 (Divisors of Meromorphic Functions)
Let X be a complex manifold. Let $f \in \mathcal{M}_X^*$ be a meromorphic function. Define the divisor of f to be

$$\text{div}(f) = \sum_V \nu_V(f) \cdot V$$

where ν_V is the valuation of $\mathcal{O}_{X,V}$.

Definition 18.1.5 (Cartier Divisors on a Complex Manifold)
Let X be a complex manifold. Define the group of divisors of X to be

$$\text{CDiv}(X) = H^0(X, \mathcal{M}_X^*/\mathcal{O}_X^*) = \Gamma(X, \mathcal{M}_X^*/\mathcal{O}_X^*)$$

Proposition 18.1.6
Let X be a complex manifold. Then there is a natural isomorphism

$$\text{Div}(X) \cong \text{CDiv}(X)$$

18.2 Holomorphic Line Bundles

Proposition 18.2.1
Let X be a complex manifold. Then the following diagram is commutative:

$$\begin{tikzcd} \text{CDiv}(X) \arrow[r] \arrow[d, "\cong"'] & \text{Pic}(X) \arrow[d, "\cong"] \\ H^0(X, Q_X^*/\mathcal{O}_X^*) \arrow[r] & H^1(X, \mathcal{O}_X^*) \end{tikzcd}$$

where the bottom horizontal map is induced by the long exact sequence of the short exact sequence $0 \to \mathcal{O}_X^* \to \mathcal{M}_X^* \to \frac{\mathcal{M}_X^*}{\mathcal{O}_X^*} \to 0$.

By Serre's correspondence, holomorphic line bundles correspond to locally free sheaves of $\mathcal{O}_M$-modules of rank 1, and recall isomorphism classes of these objects form a group structure with the tensor product, and it is group isomorphic to $H^1(M, \mathcal{O}_M^*)$.

Definition 18.2.2 (The Exponential Sequence)
Let M be a complex manifold. Define the exponential sequence of M to be the short exact sequence of sheaves of $\mathcal{O}_M$-modules given by

$$0 \longrightarrow \underline{\mathbb{Z}} \hookrightarrow \mathcal{O}_M \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_M^* \longrightarrow 0$$

Proposition 18.2.3
Let M be a complex manifold. Let L be a holomorphic line bundle on M. Then the first chern class of L is given by the image of L in the homomorphism

$$\text{Pic}(M) \cong H^1(M, \mathcal{O}_M^*) \to H^2(M, \underline{\mathbb{Z}}) \cong H^2(M, \mathbb{Z})$$

induced by the long exact sequence of the exponential sequence.

18.3 Kodaira Dimension

Definition 18.3.1
Let M be a complex manifold. Let L be a holomorphic line bundle on M. Define ??? to be

$$R(M, L) = \bigoplus_{m=0}^{\infty} H^0(M, L^{\otimes m})$$

Definition 18.3.2 (The Canonical Ring)
Let M be a complex manifold. Define the canonical ring of M to be

$$R(M) = R(M, \mathcal{M}_M) = \bigoplus_{m=0}^{\infty} H^0(M, \mathcal{M}_M^{\otimes m})$$

Lemma 18.3.3
Let M be a complex manifold. If M is connected, then $R(M)$ is an integral domain.

Definition 18.3.4 (Kodaira Dimension)
Let M be a connected complex manifold. Define the Kodaira dimension of M to be

$$\text{kod}(M) = \begin{cases} -\infty & \text{if } R(X) = \mathbb{C} \\ \text{trdeg}_{\mathbb{C}}(Q(R(X))) - 1 & \text{otherwise} \end{cases}$$

19 Compact Kahler Manifolds

19.1 Kodaira Embedding Theorem

Proposition 19.1.1
Let X be a smooth projective variety over $\mathbb{C}$. Let $\mathcal{L}$ be a line bundle over X. Then $\mathcal{L}$ is ample if and only if $\mathcal{L}^{\text{an}}$ is positive.

Theorem 19.1.2 (Kodaira Embedding)
Let X be a compact complex manifold. Then the following are equivalent.
- X admits a holomorphic embedding into $\mathbb{CP}^n$.
- X admits an ample line bundle.
- X admits a positive line bundle.
- X is admits a Kahler metric ω such that $[\omega] \in H^{1,1}(X, \mathbb{Z})$.
- There exists a projective variety Z over $\mathbb{C}$ such that $Z^{\text{an}} = X$.

Corollary 19.1.3
There is an equivalence of categories between smooth projective varieties over $\mathbb{C}$ and compact Kahler manifolds given by the analytification functor.

Theorem 19.1.4 (Kodaira-Nakano Vanishing Theorem)
Let X be a compact Kahler manifold. Let $p : L \to X$ be a positive line bundle. Then we have

$$H^q(X, \Omega^p(L)) = 0$$

for $p + q > \dim(X)$.

Theorem 19.1.5 (Lefschetz Hyperplane Theorem)
Let X be a compact Kahler manifold. Let $V \subseteq X$ be a smooth variety of dimension $\dim(X) - 1$. Then the restriction map

$$H^k(X, \mathbb{Z}) \to H^k(V, \mathbb{Z})$$

induces isomorphisms for $0 \leq k < \dim(X) - 1$ and is injective for $k = \dim(X) - 1$.

Theorem 19.1.6 (The Lefschetz $(1, 1)$ Theorem)
Let X be a compact Kahler manifold. Let $\alpha \in H^2(X, \mathbb{Z})$. Then α is the first Chern class of a line bundle if and only if $\alpha \in H^{1,1}(X)$.

19.2 Projective Curves

Proposition 19.2.1
Let X be a smooth projective curve over $\mathbb{C}$. Then X^{an} is homeomorphic to $\Sigma_{p_g(X)}$ the orientable surface of genus $p_g(X)$.

Corollary 19.2.2
There is an equivalence of categories between smooth projective curves over $\mathbb{C}$ and compact Riemann surfaces.

20 Analytic Manifolds

20.1 Real Analytic Manifolds

Definition 20.1.1 (Real Analytic Manifolds) A real analytic manifold is a topological manifold with analytic transition maps.

In the real case, every analytic manifold is a smooth manifold because every analytic function is necessarily infinitely differentiable. While not every analytic function is smooth, we can still equip a smooth structure on analytic manifolds.

Theorem 20.1.2 ((Grauert-Whitney)) Every smooth manifolds admits a compatible real analytic structure.

Definition 20.1.3 (Analytic Functions) Let M be a real analytic manifold. Let $U \subseteq M$ be a subset. Let $f : U \to \mathbb{R}$ be a function. We say that f is analytic if for all $x \in U$, there exists a chart $(V, \varphi = x^1, \dots, x^n))$ of M such that

$$f \circ \varphi^{-1} : \mathbb{R}^n \to \mathbb{R}$$

is analytic in the usual sense.

Definition 20.1.4 (Sheaf of Analytic Functions) Let M be an analytic manifold. Define the sheaf of analytic functions

$$\mathcal{A}_M : \textbf{Open} \to \textbf{CRing}$$

as follows.

- For each open set $U \subseteq M$, $\mathcal{A}_M(U)$ consists of the ring of analytic functions on U
- For each inclusion $V \subseteq U$, there is a unique morphism

  $$\text{res}^U_V : \mathcal{A}_M(U) \to \mathcal{A}_M(V)$$

  given by $f \mapsto f|_V$

21 Analytic Varieties

21.1 Real Analytic Varieties

21.2 Regular and Singular Points

Definition 21.2.1 (Regular and Singular Points) Let M be a complex manifold and let A be an analytic subset of M. We say that $x \in A$ is a regular point if there exists some open neighbourhood U of x such that $A \cap U$ is a complex submanifold of M. Otherwise, x is said to be singular. Denote

$$A_{\text{reg}} = \{x \in A \mid x \text{ is a regular point }\} \quad \text{and} \quad A_{\text{sing}} = \{x \in A \mid x \text{ is a singular point }\}$$

Theorem 21.2.2 Let M be a complex manifold and let A be an analytic subset of M. Then A_{sing} is an analytic subset of A.

22 Intersection Theory

22.1 The Divisor of a Rational Function

Definition 22.1.1 (The Order of a Function at a Subvariety)
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f/g \in K(W)^*$ for $f, g \in \mathcal{O}_X(W)$ and $g \neq 0$. Let $V \subseteq W$ be an irreducible subvariety of codimension 1. Define the order of f at V to be

$$\text{ord}_V(f/g) = l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(f)}\right) - l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(g)}\right)$$

Definition 22.1.2 (The Divisor of a Rational Function)
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f \in K(W)^*$. Define the divisor of f to be

$$\text{div}(f) = \sum_{\substack{V \text{ is a codimension 1} \\ \text{irreducible subvariety of } W}} \text{ord}_V(f) \cdot V$$

22.2 The Chow Groups

Definition 22.2.1 (The Group of n-Cycles)
Let k be a field. Let X be a scheme of finite type over k. Let $n \in \mathbb{N}$. Define the group of n-cycles on X to be the free abelian group

$$Z_n(X) = \mathbb{Z}\langle V \mid {}_{\text{irreducible subvariety of } X}^{V \text{ is an } n\text{-dimensional}} \rangle$$

Lemma 22.2.2
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f \in K(W)^*$. Then $\text{div}(f) \in Z_{\dim(W)-1}(X)$.

Definition 22.2.3 (Cycles Rationally Equivalent to Zero)
Let k be a field. Let X be a scheme of finite type over k. Let $V \in Z_n(X)$. We say that V is rationally equivalent to 0 if there exists finitely many $n+1$-dimensional irreducible subvarieties $W_1, \dots, W_r$ of X and $f_i \in K(W_i)^*$ such that

$$V = \sum_{i=1}^{n} \text{div}(f_i)$$

Proposition 22.2.4
Let k be a field. Let X be a scheme of finite type over k. Let $V \in Z_n(X)$ be a cycle. Then V is rationally equivalent to 0 i and only if there exists $(n+1)$-dimensional irreducible subvarieties $W_1, \dots, W_r$ of $X \times \mathbb{P}^1_k$ such that the projection maps $p_i : W_i \to \mathbb{P}^1_k$ are dominant and $V = \sum_{i=1}^{r}[p_i^{-1}(0)] - [p_i^{-1}(\infty)]$

Definition 22.2.5 (The Chow Group)
Let k be a field. Let X be a scheme of finite type over k. Define the nth Chow group of X to be

$$\text{CH}_n(X) = \frac{Z_n(X)}{\sim}$$

where we say that $V_1 \sim V_2$ if $V_1 - V_2$ is rationally equivalent to 0. In this case, we say that V_1 and V_2 are rationally equivalent.

22.3 The Ring Structure

22.4 Pushfoward and Pullback of Cycles

Definition 22.4.1 (The Push Forward Map)
Let k be a field. Let X, Y be schemes of finite type over k. Define the push forward map $f_* : Z_n(X) \to Z_n(Y)$ by
$$f_*(V) = \deg(V/f(V))f(V)$$
where
$$\deg(V/f(V)) = \begin{cases} [k(V) : k(f(V))] & \text{if } \dim(V) = \dim(f(V)) \\ 0 & \text{otherwise} \end{cases}$$

Proposition 22.4.2
Let k be a field. Let X, Y be schemes of finite type over k. Then the push forward map descends to a well defined homomorphism $f_* : \mathrm{CH}_n(X) \to \mathrm{CH}_n(Y)$.

Proposition 22.4.3
Let k be a field. Let X be a scheme of finite type over k. Let $i : Y \hookrightarrow X$ be a closed subscheme of X. Let $j : X \setminus Y \hookrightarrow X$ be the inclusion map. Then there is an exact sequence
$$\mathrm{CH}_n(Y) \xrightarrow{i_*} \mathrm{CH}_n(X) \xrightarrow{j^*} \mathrm{CH}_n(X \setminus Y) \longrightarrow 0$$

23 Algebraic de Rham Cohomology

Definition 23.0.1 (Algebraic de Rham Cohomology)
Let S be a scheme. Let X be a scheme over S. Define the nth algebraic de Rham cohomology group by
$$H^n_{\mathrm{dR}}(X/S) = \mathbb{H}^n(X; \Omega^\bullet_{X/S})$$

Lemma 23.0.2
Let S be a scheme. Let X be a scheme over S. Then $\mathrm{Tot}^\oplus\left(\mathcal{C}^\bullet(X, \mathcal{U}, \Omega^\bullet_{X/S})\right)$ is an acyclic resolution of $\Omega^\bullet_{X/S}$.

Thus one way to compute the de Rham cohomology would be
$$H^n_{\mathrm{dR}}(X/S) \cong H^n(\mathrm{Tot}^\oplus(\mathcal{C}^\bullet(X, \mathcal{U}, \Omega^\bullet_{X/S}))$$

Example 23.0.3
Let $U_0 = \mathrm{Spec}(R_0 = k[x,y]/(y^2 - f(x)))$ where k is a field of characteristic 0 and f is separable and $\deg(f) = 2g+1$ or $2g+2$. Let $U_1 = \mathrm{Spec}(R_1 = k[u,v]/(v^2 - h(u)))$ where $h(u) = u^{2g+2}h(1/u)$. Then we have
$$H^i_{\mathrm{dR}}(X/k) = H^i\left(R_0 \oplus R_1 \to R_0\frac{dx}{y} \oplus R_1\frac{du}{v} \oplus R_0[1/x] \to R_0[1/x]dx\right)$$
where the first map is given by $(w, z) \mapsto (dw, dz, w - z(1/x, y/x^{g+1}))$, the second map is given by $(wdx, zdu, g(x,y)) \mapsto wdx - zdu - dg$ where $du = -z(1/x, y/x^{g+1})dx/x^2$.

- $H^0_{\mathrm{dR}}(X/k) \cong k$.
- $H^2_{\mathrm{dR}}(X/k) \cong H^1_{\mathrm{dR}}(X/k)$.
- $H^0(X, \Omega^0_{X/k}) \hookrightarrow H^1_{\mathrm{dR}}(X/k)$ given by $\omega \mapsto (\omega|_{U_0}, \omega|_{U_1}, 0)$ is an inclusion since the global sections are constant.
- There is a map $H^1_{\mathrm{dR}}(X/kk) \to H^1(X, \mathcal{O}_X$ given by $(\omega_0, \omega_1, g) \mapsto g$ and precomposing with the above map gives 0. It is clearly surjective.

- So we have an exact sequence
$$0 \to H^0(X, \Omega^\bullet_{X/k}) \to H^1_{\mathrm{dR}}(X/k) \to H^1(X, \mathcal{O}_X) \to 0$$
called the Hodge filtration on $H^1_{\mathrm{dR}}(X/k)$. In characteristic p this is more subtle: not all global differentials are closed.
- What can we say about the map $H^1_{\mathrm{dR}}(X/k) \to H^1_{\mathrm{dR}}(U_0/k)$ given by $(\omega_0, \omega_1, g) \mapsto g$.
- The LES in local cohomology gives
$$0 \to H^1(X(\mathbb{C}), \mathbb{C}) \to H^1(U_0(\mathbb{C}), \mathbb{C}) \to H^2(X(\mathbb{C}), U_0(\mathbb{C}), \mathbb{C}) \to H^2(X(\mathbb{C}), \mathbb{C}) \to 0$$
and $H^2(X(\mathbb{C}), U_0(\mathbb{C}), \mathbb{C}) \cong H^0((X - U_0)(\mathbb{C}), \mathbb{C}) \cong \mathbb{C}^{|X - U_0|(\mathbb{C})}$. If $\deg(f)$ is odd, $|(X - U_0)(\mathbb{C})| = 1$. Then $H^1(X(\mathbb{C}), \mathbb{C}) \to H^1(U_0(\mathbb{C}), \mathbb{C})$ is an isomrophism. If $\deg(f)$ is even, then we have $0 \to H^1(X(\mathbb{C}), \mathbb{C}) \to H^1(U_0(\mathbb{C}), \mathbb{C}) \to \mathbb{C} \to 0$. Same thing happens for de Rham cohomology. Surjectivity of $H^1_{\mathrm{dR}}(X/k) \to H^1_{\mathrm{dR}}(U_0/k)$ iff every class (ω_0) for $\omega_0 \in \Omega^1_{R_0/k}$ is the image of $\ker(R_0 dx/y \oplus R_1 du/v \oplus R_0[1/x] \to R_0[1/x]dx)$. i.e. $\exists \omega_1 \in R_1 du/v$, $g \in R_0[1/x]$ such that $\omega_0 - \omega_1 = dg$. The obstruction to finding such an ω_1 and g would be the residue of a differential. Pick $z \in (X - U_0)(\overline{k})$ a closed point and a parameter t at z. Write ω as $F(t) = \sum_{i=0}^\infty a_i t^i \in \overline{k}[[t]]$. Eg any meromorphic can be written as rational function times dt, then, take formal power series. Then $\mathrm{Res}_z(\omega) = a_1$ of $F(t)$. If $\omega_1 \in \Omega^1_{R_1/k}$ then $\mathrm{Res}_z = 0$ (indeed it has no pole). Also $\mathrm{Res}_z(dg) = 0$. Then image of $H^1_{\mathrm{dR}}(X/k)$ is contained in the kernel of the residue map
$$\mathrm{Res} : H^0_{\mathrm{dR}}(U_0/k) \to \bigoplus_{Z \in (X - U_0(\overline{k})} \overline{k}$$
Fact: For any open $U \subseteq X$, $\omega \in H^0(U, \Omega_{X/k})$, then
$$\sum_{Z \in (X-U)(\overline{k})} \mathrm{Res}_z(\omega) = 0$$

Proposition 23.0.4
Let S be a scheme. Let X be a scheme over S. Then the following are true.
- $H^n_{\mathrm{dR}}(X/S)$ is a $H^0_{\mathrm{dR}}(X/S)$-module.
- The map $H^i_{\mathrm{dR}}(X/S) \times H^j_{\mathrm{dR}}(X/S) \to H^{i+j}_{\mathrm{dR}}(X/S)$ induced by the wedge of differential forms is a $H^0_{\mathrm{dR}}(X/S)$-linear.

Definition 23.0.5 (The de Rham Cohomology Ring)
Let S be a scheme. Let X be a scheme over S. Define the de Rham cohomology ring of X to be
$$H^*_{\mathrm{dR}}(X/S) = \bigoplus_{i=0}^\infty H^i_{\mathrm{dR}}(X/S)$$
together with ring structure given by wedge of differential forms.

Lemma 23.0.6
Let S be a scheme. Let X be a scheme over S. Then $H^*_{\mathrm{dR}}(X/S)$ is a graded commutative $H^0_{\mathrm{dR}}(X/S)$-algebra.

Lemma 23.0.7
Let k be a field of characteristic 0. Let X be a smooth affine variety over k. Then we have
$$H^n_{\mathrm{dR}}(X) = H^n(\Gamma(X, \Omega^\bullet_{X/k}))$$
for all $n \in \mathbb{N}$.

Example 23.0.8
Let k be a field of characteristic 0. Let $R = k[x_1, \dots, x_n]$ be the polynomial ring over k. Then $\Gamma\left(X, \Omega^1_{\text{Spec}(R)/k}\right) = \Omega^1_{R/k} = \bigoplus_{i=1}^n R dx_i$. Then we have

$$\Gamma\left(X, \Omega^j_{\text{Spec}(R)/k}\right) = \bigoplus_{1 \leq i_1 < \cdots < i_j \leq n} R dx_{i_1} \wedge \cdots \wedge dx_{i_j}$$

Then we have

$$H^i_{\text{dR}}(R/k) = \begin{cases} k & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

TBR:

Example 23.0.9
Let k be a field of characteristic $p > 0$. Then we have

$$\dim_k(H^1(\Gamma(\text{Spec}(k[t]), \Omega^1_{\text{Spec}(k[t])/k}))) = \infty$$

Indeed, we have

$$H^1(\Gamma(\text{Spec}(k[t]), \Omega^1_{\text{Spec}(k[t])/k})) = \frac{k[t]}{\langle f'(t) \mid f \in k[t] \rangle} = \frac{k[t]}{\bigoplus_{i=1}^\infty it^{i-1} dt} \cong \bigoplus_{i=1}^\infty t^{ip-1} dt$$

Example 23.0.10
$\Omega^1_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}} \cong (\mathbb{Z}/2\mathbb{Z}) d\sqrt{2}$. $H^1(\Gamma(\text{Spec}(\mathbb{Z}\sqrt{2}), \Omega^1_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}})) \neq 0$

Example 23.0.11
Let k be a field of characteristic $p \neq 2$. Let $f \in k[x]$ be separable. Let $R = \frac{k[x,y]}{y^2 - f(x)}$. Then we have

$$\Omega^1_{R/k} \cong \frac{\Omega^1_{k[x,y]/k} \otimes_{k[x,y]} k}{(2ydy - f'(x)dx)} \cong \frac{Rdx \oplus Rdy}{R(2ydy - f'(x)dx)}$$

Using the quotient relation, we may write $dy = \frac{f'(x)dx}{2y}$ in the module. Thus dx/y is a generator of the module.

Since f is separable, there exists $g, h \in k[x]$ such that $gf + hf' = 1$. Let $\omega = aydx + 2bdy \in \Omega^1_{R/k}$. Then we have $y\omega = dx$. Hence y divides dx and $\omega = dx/y$.

Now let $z = a(x) + b(x)y \in R$ be arbitrary. Then we have

$$dz = (a'(x) + b'(x)y)dx + h(x)dy = (ya'(x) + b'(x)f(x) + \frac{1}{2}b(x)f'(x))\omega$$

Thus every element of $\Omega^1_{R/k}$ is of the form $q(x)\omega$ for some $q(x) \in k[x]$. If $\deg(q) \geq \deg(f)$, then there exists $p \in k[x]$ such that $\deg(p) > \deg(q)$ such that

$$[p(x)\omega] = [q(x)\omega]$$

in $H^1_{\text{dR}}(R/k)$.

Lemma 23.0.12
Let k be a field of characteristic 0. Let X be a smooth variety over k. Then the following are true.

- $H^i_{\text{dR}}(X) = 0$ for all $i > 2\dim(X)$.
- If X is affine, then $H^i_{\text{dR}}(X) = 0$ for all $i > \dim(X)$.

Proposition 23.0.13

23.1 Hodge de Rham Spectral Sequences

Definition 23.1.1 (The Hodge Filtration)
Let S be a scheme. Let X be a scheme over S. Define the Hodge filtration $F^\bullet\Omega^\bullet_{X/S}$ of X to be

$$F^i\Omega^j_{X/S} = \begin{cases} \Omega^j_{X/S} & \text{if } j \geq i \\ 0 & \text{otherwise} \end{cases}$$

Thus for each i, the collection of complexes

$$F^i\Omega^\bullet_{X/S} = (0 \to \Omega^i_{X/S} \to \Omega^{i+1}_{X/S} \to \cdots)$$

form the filtration for $\Omega^\bullet_{X/S}$.

Lemma 23.1.2
Let S be a scheme. Let X be a scheme over S. Then $F^\bullet\Omega^\bullet_{X/S}$ is an exhaustive filtration.

Definition 23.1.3 (The Hodge de Rham Spectral Sequence)
Let S be a scheme. Let X be a scheme over S. Define the Hodge de Rham spectral sequence to be the spectral sequence associated to the Hodge siltration $F^\bullet\Omega^\bullet_{X/S}$.

The E^0 page is given by

$$E_0^{p,q} = \frac{F^p\Omega^{p+q}_{X/S}}{F^{p+1}\Omega^{p+q}_{X/S}} = \frac{\Omega^q_{X/S}}{\Omega^{q-1}_{X/S}}$$

Lemma 23.1.4
Let S be a scheme. Let X be a scheme over S. Then we have

$$E_1^{p,q} = H^q(X, \Omega^p_{X/S}) \Rightarrow H^{p+q}_{\text{dR}}(X/S)$$

Theorem 23.1.5 (Deligne)
If $S = \text{Spec}(k)$, k char 0, $X \to S$ smooth and proper. Then the Hodge de Rham spectral sequence degenerates at the E_1 page (all differentials in page 1 is 0).

When you have a 2-step filtration: $0 \to F^1C^\bullet \to C^\bullet \to C^\bullet/F^1C^\bullet \to 0$, the spectral sequence is the long exact sequence in cohomology.

23.2 Flat Connections

$X/\mathbb{C}$ smooth connected algebraic variety. $(V, \Delta) \to (V^{\text{an}})^{\Delta=0}$.

Cyclic vector theorem: (V, Δ) on $K(C)$, C a connected curve over k. Suppose that $e \in V$. Let $e_i = \partial_i e$.

24 Deformation of Manifold Structures

24.1

Recall:

$$\begin{tikzcd} \text{Smooth Manifolds} \arrow[r, phantom, "\subset"] & \text{PL Manifolds} \arrow[r, phantom, "\subset"] \arrow[d, leftrightarrow] & \text{Topological Manifolds} \\ & \text{Combinatorial Manifolds} \arrow[r, phantom, "\subset"] & \text{Triangulated Manifolds} \arrow[u] \end{tikzcd}$$

is every triangulated manifold a combinatorial manifold? manifold Hauptvermutung:

Definition 24.1.1 (Smooth Embedding of a Simplex)
Let $\Delta^k \subseteq \mathbb{R}^{k+1}$ be the standard k-simplex. Let M be a smooth manifold. Let $\sigma : \Delta^k \to M$ be a map. We say that σ is a smooth embedding if the following are true.
- f is injective.
- f induces a homeomorphism from Δ^k to $\text{im}(f)$.
- For all facets $F \subseteq \Delta^k$, the induced map $\sigma|_{F^\circ}$ is an immersion of smooth manifolds.

Definition 24.1.2 (Smooth Triangulations)
Let M be a smooth manifold with boundary. Let K be a simplicial complex that is a triangulation of M. We say that K is a smooth triangulation if the following are true.
- For each simplex $\sigma \in K$, the map σ is a smooth embedding.
- There exists a subcomplex $K' \subseteq K$ such that K' is a triangulation of ∂M.

Lemma 24.1.3
Let M be a smooth manifold with boundary. Let K be a smooth triangulation of M. Then M admits a PL structure.

Theorem 24.1.4 (Whitehead, 1940)
Let M be a smooth manifold with boundary. Then M admits a smooth triangulation. Moreover, if K and K' are both smooth triangulations, then $|K|$ and $|K'|$ are PL-homeomorphic.

24.2 Microbundles

Define the tangent microbundle τ_M for a topological n-manifold M. Stable topological microbundles over M are classified by $BTop = \varinjlim_k B\text{Top}(k)$, so the stable tangent microbundle determines a classifying map $M \to BTop$. Since $\text{Top}(k)/\text{PL}(k) \to Top/PL$ is a homotopy equivalence for $k \geq 5$, a lift of $M \to BTop$ through $BPL \to BTop$ is equivalent to a PL structure on the unstable tangent microbundle τ_M.

Lemma 24.2.1
The following are true.
- There is a homotopy equivalence
$$\text{hofib}(BPL \to BTop) \simeq K(\mathbb{Z}/2\mathbb{Z}, 3)$$
- $BPL \to BTop$ is a principal fibration.
- The principal fibration induces a fibration $BTop \to K(\mathbb{Z}/2\mathbb{Z}, 4)$.

24.3 The Tangent Microbundle

Theorem 24.3.1 (Hirsch-Mazur Theorem)
Let M be a topological manifold of dimension $n \geq 5$. Then τ_M admits a PL structure if and only if M admits a PL structure.

24.4 The Kirby-Siebenmann Class

Definition 24.4.1 (The Kirby-Siebenmann Class)
Let M be a topological manifold. Define the Kirby-Siebenmann class of M to be the element

$$\kappa(M) \in H^4(M; \mathbb{Z}/2\mathbb{Z}) \cong [M, K(\mathbb{Z}/2\mathbb{Z}, 4)]$$

given by the element $M \overset{\tau M}{\to} BTop \to K(\mathbb{Z}/2\mathbb{Z}, 4)$.

Proposition 24.4.2 (Kirby-Siebenmann Structure Theorem)
Let M be a compact topological manifold of dimension $n \geq 5$. Then the following are true.

- M admits a PL structure if and only if $\kappa(M) = 0$.
- If $\kappa(M) = 0$, then the set of concordance classes of PL structure on M is a torsor over $H^3(M; \mathbb{Z}/2\mathbb{Z})$.

24.5 The Low Dimensional Case

Theorem 24.5.1 (Rado's Theorem)
Let M be topological manifold of dimension 2. Then M admits a unique smooth structure, unique PL structure and unique triangulation.

Theorem 24.5.2 (Moise's Theorem)
Let M be topological manifold of dimension 3. Then M admits a unique smooth structure, unique PL structure and unique triangulation.

Theorem 24.5.3 (Cairns-Hirsch)
Let M be piecewise linear manifold of dimension 4. Then M admits a unique smooth structure and a unique PL structure.

Theorem 24.5.4
Let M be a topological manifold of dimension 4. Then the following are true.

- If $\kappa(M) \neq 0$, then M admits no PL structure and no smooth structure.
- If $\kappa(M) = 0$ and Q_M is not diagonalizable over $\mathbb{Z}$, then M admits no PL structure and no smooth structure.

Theorem 24.5.5
The following are true.

- There exists uncountably many pairwise non-diffeomorphic smooth structures on $\mathbb{R}^4$.
- For $n \neq 4$, there exists a unique smooth structure on $\mathbb{R}^n$.

Theorem 24.5.6 (Manolescu 2016)
For all $n \geq 5$, there exists a compact topological manifold of dimension n that admits no triangulation.

Theorem 24.5.7
Let M be PL manifold of dimension $n \leq 6$. Then M admits a smooth structure.

$$\underset{\text{Manifolds}}{\text{Smooth}} \subset \underset{\text{Manifolds}}{\text{PL}} \subset \underset{\text{Manifolds}}{\text{Topological}}$$

Summarizing: All inclusions are equalities for $\dim \leq 3$. The first inclusion is strict for $\dim = 4$.

25 Rational Homotopy Theory

25.1 Homology with Coefficients in a Subring of $\mathbb{Q}$

Theorem 25.1.1 (Whitehead-Serre Theorem)
Let X, Y be simply connected spaces. Let R be a subring of $\mathbb{Q}$. Let $f : X \to Y$ be a map. Then the following are equivalent.
- $\pi_*(f) \otimes_{\mathbb{Z}} R$ is an isomorphism of graded abelian groups.
- $H_*(f; R)$ is an isomorphism of graded rings.
- $H_*(\Omega f; R)$ is an isomorphism of graded rings.

Lemma 25.1.2
Let X be an n-connected space. Let R be a subring of $\mathbb{Q}$. Suppose that one of the following are true.
- $n \geq 1$.
- $n = 0$ and $\pi_1(X)$ is abelian.

Then the Hurewicz homomorphism induces an isomorphism
$$\pi_{n+1}(X) \otimes_{\mathbb{Z}} R \cong H_{n+1}(X; R)$$

Proposition 25.1.3
Let A be an abelian group. Let $n \in \mathbb{N}^+$. Let R be a subring of $\mathbb{Q}$. Then we have $A \otimes_{\mathbb{Z}} R = 0$ if and only if $H_*(K(A, n); R) \cong H_*(*; R)$.

25.2 P-Local Modules

Definition 25.2.1 (P-Local Groups)
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. We say that A is P-local if for all $q \in \mathbb{N} \setminus P$, the map $\cdot q : A \to A$ is an isomorphism.

Definition 25.2.2 (P-Localization Map)
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Define the P-localization map of A to be the map
$$A \to A \otimes_{\mathbb{Z}} S_P^{-1}\mathbb{Z}$$
where $S_P = \{n \in \mathbb{Z} \mid \gcd(n, p) = 1 \text{ for all } p \in P\}$.

Proposition 25.2.3
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Let $S_P = \{n \in \mathbb{Z} \mid \gcd(n, p) = 1 \text{ for all } p \in P\}$. Then the following are equivalent.
- A is P-local.
- A is a $S_P^{-1}\mathbb{Z}$-module by extension of scalars.
- The P-localization map $A \to A \otimes_{\mathbb{Z}} S_P^{-1}\mathbb{Z}$ is an isomorphism.

When $P = \emptyset$, then $S_P^{-1}\mathbb{Z} = \mathbb{Q}$ is just rationalization.

25.3 P-Local Spaces

Definition 25.3.1 (P-Local Space)
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. We say that X is P-local if $\pi_n(X)$ is a P-local group for all $n \geq 2$.

Proposition 25.3.2
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Then the following are equivalent.

- X is P-local.
- $H_n(X,*)$ is P-local for all $n\geq 0$.
- $H_n(\Omega X,*)$ is P-local for all $n\geq 0$.

Lemma 25.3.3
Let A be an abelian group. Let $n\geq 1$. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. If A is P-local, then we have
$$H_*(X;\mathbb{Z}/p\mathbb{Z})\cong H_*(*,\mathbb{Z}/p\mathbb{Z})$$
for all $p\notin P$.

Definition 25.3.4 (P-Localization)
Let X be a simply connected space. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. A P-localization of X is a space X_P together with a map $i_p:X\to X_P$ such that f induces an isomorphism
$$\pi_n(X)\otimes_{\mathbb{Z}}S_P^{-1}\mathbb{Z}\to\pi_n(X_P)$$
for all $n\geq 2$.

Proposition 25.3.5
Let X be a simply connected space. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. Suppose that X_P exists. Then the following are true.
- Suppose that Z is a P-local space and $f:X\to Z$ is a map. Then there exists a map $g:X_P\to Z$ such that $g\circ i_p=f$.
- Suppose that Z is a P-local space and $f_1,f_2:X\to Z$ are map. Suppose that $g_1,g_2:X_P\to Z$ are maps such that $g_1\circ i_p=f_1$ and $g_2\circ i_p=f_2$. Suppose that $H:X\times I\to Z$ is a homotopy from f_1 to f_2. Then there exists a homotopy $K:X_P\times I\to Z$ such that $K\circ i_p\times\mathrm{id}_Z=H$.

Proposition 25.3.6
Let X be a simply connected space. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. Then X_P exists and is unique up to homotopy equivalence relative to X.

25.4 Rational Homotopy Equivalence

Definition 25.4.1 (Rational Model)
Let X be a simply connected space. A rational model of X is a P-localization of X where $P=\emptyset$. It is denoted by $X_{\mathbb{Q}}$.

Proposition 25.4.2
Let X be a simply connected space. Then the following are true.
- $X_{\mathbb{Q}}$ is homotopy equivalent to a CW complex.
- $C_*(X_{\mathbb{Q}})$ has all differentials as 0.
- $H_*(X_{\mathbb{Q}})$ is a free graded $\mathbb{Z}$-module.

26 Topics in Simplicial Homology

26.1

Proposition 26.1.1
Let S be a simplicial complex. Let $\sigma \in S$. Let $p \in |\sigma|^\circ$. Then we have

$$H^k_{\text{sing}}(|S|, |S| \setminus p) \cong \widetilde{H}^{k-|\sigma|}_\Delta(\text{lk}(\sigma))$$

27 The Nature of Mathematical Physics

In mathematical physics, physical observations are given a mathematical formalism. A recurring theme is that mathematics often predict new phenomena beyond the original observation.

Mechanics

Mechanics is the part of physics that study the motion of bodies. Statics is the subject that studies equilibriums. Kinematics is concerned with general properties of motions regardless of the source. Dynamics study motion caused by forces or interactions. In regards to the system that is studied, mechanics can also be divided into material points, rigid bodies, elastic bodies and fluids.

Mechanics have received a number of revolutionary ideas that transformed the subject. Completely. In the modern day there are three main branches of mechanics: classical mechanics, relativistic mechanics and quantum mechanics.

Classical Mechanics

Theoretically, there are three different formalisms of classical mechanics that are related: Newtonian, Lagrangian and Hamiltonian. Newtonian mechanics is the easiest to formulate. Lagrangian is based on the least action principle, which is more sophisticated but is efficient for mechanical systems that are under constraints. Hamiltonian mechanics connects the bridge between classical mechanics and quantum mechanics.

For a mathematical formalism of classical mechanics, we look for a mathematical model that describes the following physically observed phenomena.

- Our space is three dimensional and time is one dimensional.
- There exists coordinate systems called inertial such that all laws of nature at all moments of time are the same in all inertial coordinates, and all coordinate systems in uniform rectilinear motion with respect to an inertial one are themselves inertial.
- The initial state of a mechanical system uniquely determines all of its motion.

The limitations of classical mechanics come from Maxwell's equations, which says that electromagnetic waves travel at a fixed speed regardless of which inertial frame one uses. In particular, classical mechanics is a good enough estimate for motions of bodies that are significantly less than the speed of light, and whose bodies are macroscopic objects.

Relativistic Mechanics

For a mathematical formalism of relativistic mechanics, we look for a mathematical model that describes the following physically observed phenomena.

- The Principle of Relativity: The laws of physics are the same in all inertial frames.
- The speed of light is the same for all inertial frames, regardless of the motion of the source or observer.

A number of consequences appear because the speed of light being constant.

- Time dilation: Moving clocks run slower.
- Length contraction: Moving objects are shortened along the direction of motion.
- Relativity of simultaneity: Two events simultaneous in one frame are not simultaneous in another.
- Speed do not add linearly, nothing with mass can have speed that is the speed of light.
- Mass energy equivalence: $E = mc^2$.
- Instantaneous interactions are impossible, and so must propagate at finite speed.

Special relativity introduces the framework that is Lorentz geometry for the universe. This framework is one that is compatible with the observation that is the speed of light is the same for all inertial frames. In particular, it deduces the relativity of simultaneity, which means that instantaneous interactions are impossible. This introduces the concept of fields and field theory.

The limitations of relativistic mechanics come from general relativity, which says that gravity should be a result of the geometry, not a force. Relativistic mechanics does not take this to account. Also, classical field theory cannot handle the destruction and creation of particles.

General Relativity

Quantum Mechanics

Quantum Field Theory

Statistical Mechanics

Thermal Dynamics

Quantum Information

Condensed Matter Physics

Cosmology

28 K Theory

28.1

Definition 28.1.1 (The 0th K-Group)
Let R be a ring. Define the 0th K-group of R to be the set

$$K_0(R) = \frac{\{P \mid P \text{ is a finitely generated projective } R\text{-module }\}}{\cong}$$

together with addition given by $[P_1] + [P_2] = [P_3]$ if there is a short exact sequence
$$0 \longrightarrow P_1 \longrightarrow P_3 \longrightarrow P_2 \longrightarrow 0$$

Example 28.1.2
Suppose that R is a PID. Then there is an isomorphism

$$K_0(R) \cong \mathbb{Z}$$

given by $[n] \mapsto [R^n]$.

Definition 28.1.3 (The Change of Rings Homomorphism)
Let R, S be rings. Let $f : R \to S$ be a ring homomrphism. Define the change of rings homomorphism to be the map $K_0(f) : K_0(R) \to K_0(S)$ given by

$$[P] \mapsto [S \otimes_R P]$$

Recall the Grothendieck completion of an abelian semi-group $G(-)$.

Proposition 28.1.4
Let R be a ring. Let $P(R) = \{P \mid P \text{ is a finitely generated projective } R\text{-modules }\}$. Then there is a natural isomorphism
$$K_0(R) \cong G(P(R), \oplus)$$

Proposition 28.1.5 (Morita Equivalence)
Let R be a ring. Then there is a natural isomorphism

$$K_0(R) \cong K_0(M_n(R))$$

given by $[P] \mapsto R^n \otimes_R P$ where R^n is considered as an $(M_n(R), R)$-bimodule.

Proposition 28.1.6
Let R, S be rings. Then there is a natural isomrophism

$$K_0(R \times S) \cong K_0(R) \times K_0(S)$$

induced by the product of the projection maps.

Example 28.1.7
Let F be a field. Let V be a vector space over F of countable dimension. Then there is an isomorphism
$$K_0(\text{End}_F(V)) \cong \{1\}$$

Definition 28.1.8 (The Reduced 0th K-Group)

Let R be a ring. Define the reduced 0th K-group to be the quotient

$$\widetilde{K}_0(R) = \frac{K_0(R)}{\langle [R^m] - [R^n] \mid m, n \in \mathbb{N} \rangle}$$

Lemma 28.1.9
Let R be a ring. Then there is a natural isomorphism

$$\widetilde{K}_0(R) \cong \langle [R] \rangle \leq \widetilde{K}_0(R)$$

Proposition 28.1.10
Let G be a group. Let R be an integral domain. Then there is an isomorphism

$$K_0(R[G]) \cong \widetilde{K}_0(R[G]) \oplus \mathbb{Z}$$

give by $[P] \mapsto ([P], \dim_{\text{Frac}(R)}(\text{Frac}(R) \otimes_{R[G]} P))$

Conjecture 28.1.11 (Farrell-Jones Conjecture)
Let R be a regular ring. Let G be a torsion free group. Then the inclusion map induces a natural isomorphism

$$K_0(R) \cong K_0(R[G])$$

29 Ergodic Theory

29.1

Definition 29.1.1 (Measure Preserving Systems)
Let $(X, \mathcal{F}, \mu)$ be a measure space. Let $T : X \to X$ be a measurable function. We say that T is a measure preserving function if

$$\mu(T^{-1}(E)) = \mu(E)$$

for all $E \in \mathcal{F}$. In this case, we call $(X, \mathcal{F}, \mu, T)$ a measure preserving system.

Definition 29.1.2 (Probability Preserving System)
Let $(X, \mathcal{F}, \mu, T)$ be a measure preserving system. We say that it is a probability preserving system if $(X, \mathcal{F}, \mu)$ is a probability space.

Theorem 29.1.3 (Poincare Recurrence Theorem)
Let $(X, \mathcal{F}, \mu, T)$ be a probability preserving system. Let $E \in \mathcal{F}$. Then we have

$$\mu(\{x \in E \mid \exists N \text{ such that } T^n(x) \notin E \text{ for all } n > N\}) = 0$$

In other words, most points in E go back to E infinitely often after applying a certain number of T's.

Definition 29.1.4 (Measure Theoretically Isomorphic)
Let $(X, \mathcal{F}, \mu, T)$ and $(Y, \mathcal{E}, \sigma, S)$ be measure preserving systems. Let $\pi : X \to Y$ be a measurable function. We say that π is a measure theoretically isomorphism if the following are true.

- There exists $A \subseteq X$ and $B \subseteq Y$ such that $\mu(X \setminus A) = \mu(Y \setminus B) = 0$ and $\pi|_A : A \to B$ is invertible with a measurable inverse.
- For all $E \in \mathcal{E}$, we have $\mu(\pi^{-1}(E)) = \sigma(E)$.
- $S \circ \pi = \pi \circ T$.

29.2

Definition 29.2.1 (Invariant Sets)
Let $(X, \mathcal{F}, \mu, T)$ be a measure preserving system. Let $E \in \mathcal{F}$. We say that E is invariant if

$$T^{-1}(E) = E$$

Definition 29.2.2 (Ergodic Systems)
Let $(X, \mathcal{F}, \mu, T)$ be a measure preserving system. We say that the system is ergodic if for all invariant sets E, we have $\mu(E) = 0$ or $\mu(X \setminus E) = 0$. In this case, we say that μ is an ergodic measure.

30 Classical Field Theory

Definition 30.0.1 (Classical Fields)
A classical field is an element
$$s \in \mathcal{C}^\infty(E)$$
where $E \to M$ is a vector bundle.

30.1 Lagrangian Density

Definition 30.1.1 (Lagrangian Density)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. A Lagrangian density on E is a smooth bundle map
$$\mathcal{L} : J^1(E) \to \text{Den}(M)$$

Definition 30.1.2 (Lagrangian Function)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian density. Define the Lagrangian function of $\mathcal{L}$ to be the unique function $L \in \mathcal{C}^\infty(J^1(E))$ such that
$$\mathcal{L} = L\mu_\eta$$

Recall: $j^1\phi : M \to J^1E$ is the jet prolongation of ϕ.

Definition 30.1.3 (Action Functional)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian. Let $\Omega \subseteq M$ be compact. Define the action functional of $\mathcal{L}$ on Ω to be the map $S : \mathcal{C}^\infty(E) \to \mathbb{R}$ given by
$$S[\phi] = \int_\Omega \mathcal{L}(j^1\phi)$$

Definition 30.1.4 (Admissible Variations)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\Omega \subseteq M$ be compact. Define the set of admissible variations on Ω to be the set
$$\mathcal{C}^\infty_{\partial\Omega}(E|_\Omega) = \{\delta\phi \in \mathcal{C}^\infty(E|_\Omega) \mid \delta\phi|_{\partial\Omega} = 0\})$$

Definition 30.1.5 (First Variation)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian. Let $\Omega \subseteq M$ be compact. Let $\phi \in \mathcal{C}^\infty(E)$. Let $\delta\phi \in \mathcal{C}^\infty_{\partial\Omega}(E|_\Omega)$ be an admissible variation. Define the first variation of ϕ along $\delta\phi$ by
$$\delta S[\phi, \delta\phi] = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} S[\phi + \varepsilon\delta\phi] \in \mathbb{R}$$

Definition 30.1.6 (The Euler-Lagrange Operator)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian. Define the Euler-Lagrange operator $\text{EL} : \mathcal{C}^\infty(E) \to \mathcal{C}^\infty(\text{Hom}(E, \text{Den}(M)))$ as follows. Let $(U, x^1, \cdots, x^4)$ be a chart of M. Let $(p^{-1}(U), x^i, y^j)$ be a local trivialization chart of E. Let $(p^{-1}(U), x^i, y^j, y^j_i)$ be a local chart on $J^1(E)$ where $y^j_i \in \mathcal{C}^\infty(J^1(E))$ is such that $y^j_i(j^1_p\phi) = \frac{\partial y^j \circ \phi}{\partial x^i}\Big|_p$. Then locally at the chart, the operator $\text{EL}(\phi)_p : E_p \to \text{Den}_p(M)$ is given by
$$\text{EL}(\phi)_p(v) = \sum_{j=1}^{\text{rank}(E)} y^j(v)\left(\frac{\partial L}{\partial y^j} \circ j^1\phi - \sum_{i=1}^{4} \frac{\partial}{\partial x^i}\left(\frac{\partial L}{\partial y^j_i} \circ j^1\phi\right)\right)\bigg|_p \mu_\eta$$

where $\phi \in \mathcal{C}^\infty(E)$ and $v \in E_p$.

To explain:

- For $v \in E_p$, $y^j(v)$ picks out the jth coordinate of v according to our local chart.
- $\frac{\partial L}{\partial y^j}$ is a smooth function of $J^1(E)$. We may compose with $j^1\phi : M \to J^1(E)$ to obtain a smooth function from M to $\mathbb{R}$.
- Similarly, $\frac{\partial L}{\partial y_i^j} \circ j^1\phi$ is a smooth function from M to $\mathbb{R}$. Using the local chart of M, we may differentiate this function with respect to x^i to obtain a another smooth function from M to $\mathbb{R}$.
- Notice that everything before the Minkowski density is indeed a real number. Hence $\text{EL}(\phi)_p(v)$ is indeed an element of $\text{Den}_p(M) \cong \mathbb{R}$.

Theorem 30.1.7 (Euler-Lagrange Theorem)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian. Let $\phi \in \mathcal{C}^\infty(E)$. Then we have

$$\text{EL}(\phi) = 0$$

if and only if $\delta S[\phi, \delta\phi] = 0$ for all $\delta\phi \in \mathcal{C}^\infty_{\partial\Omega}(E|_\Omega)$.

30.2 Hamiltonian Density

Definition 30.2.1 (Conjugate Momentum Density)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian. Let $t \in \mathbb{R}$. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be an inertial foliation. Let $(p^{-1}(U), x^i, y^j, y_i^j)$ be a local chart of $J^1(E)$. Define the conjugate momentum density $\pi_t : \mathcal{C}^\infty(E) \to \mathcal{C}^\infty(\text{Hom}(E|_{\Sigma_t}, \text{Den}(\Sigma_t)))$ of $\mathcal{L}$ at t at the local chart by

$$\pi_t(\phi)_p(v) = \sum_{j=1}^{\text{rank}(E)} y^j(v) \left(\frac{\partial L}{\partial y_1^j} \circ j^1\phi \right)\Bigg|_p \mu_\eta|_{\Sigma_t}$$

Definition 30.2.2 (Hamiltonian Density)
Let $p : E \to M$ be a smooth vector bundle over Minkowski spacetime. Let $\mathcal{L} : J^1(E) \to \text{Den}(M)$ be a Lagrangian. Let $t \in \mathbb{R}$. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be an inertial foliation. Let u_p be the corresponding inertial observer field. Define the Hamiltonian density of $\mathcal{L}$ to be the function $\mathcal{H}_t : \mathcal{C}^\infty(E|_{\Sigma_t}) \times \mathcal{C}^\infty(\text{Hom}(E|_{\Sigma_t}, \text{Den}(\Sigma_t))) \to \mathcal{C}^\infty(\text{Den}(\Sigma_t))$ given by

$$\mathcal{H}_t(\phi)_p = \left(\pi_t(\phi)_p(j^1\phi(p)(u_p)) - L \circ j^1\phi(p)\right) \mu_\eta|_{\Sigma_t}$$

Definition 30.2.3 (The Hamilton Operator)

Theorem 30.2.4 (Hamilton's Equations)

30.3 Covariant Hamiltonian Formalism

30.4 Scalar Fields and Klein-Gordon Lagrangian

Definition 30.4.1 (Klein-Gordon Lagrangian Density)
Define the Klein-Gordon Lagrangian density to be the bundle map $\mathcal{L}_{\text{KD}} : J^1(M \times \mathbb{R}) \to \text{Den}(M)$

given at local chart $(U, x^1, \cdots, x^4, y, y^1, \ldots, y^4)$ by

$$\mathcal{L} = \left(\frac{1}{2} \sum_{i=1}^{4} \sum_{j=1}^{4} \eta_{ij} y_i y_j - \frac{1}{2} m^2 y^2 \right) \mu_\eta$$

where (η_{ij}) is the matrix representation of η and $m \in \mathbb{R}$.

Lemma 30.4.2
Let $(U, x^1, \cdots, x^4, y, y^1, \ldots, y^4)$ be a local chart of $J^1(M \times \mathbb{R})$. Then we have

$$\mathcal{L}_{\text{KD}} = \left(\frac{1}{2} \left((y_1)^2 - (y_2)^2 - (y_3)^2 - (y_4)^2 \right) - \frac{1}{2} m^2 y^2 \right) \mu_\eta$$

Lemma 30.4.3
Let $(U, x^1, \cdots, x^4, y, y^1, \ldots, y^4)$ be a local chart of $J^1(M \times \mathbb{R})$. Then we have

$$\text{EL}(\phi)_p(v) = y(v) \left(-m^2 \phi(p) - \left(\frac{\partial^2 \phi}{\partial x_1^2} - \frac{\partial^2 \phi}{\partial x_2^2} - \frac{\partial^2 \phi}{\partial x_3^2} - \frac{\partial^2 \phi}{\partial x_4^2} \right) \right) \mu_\eta$$

Definition 30.4.4 (d'Alembertian)
Define the d'Alembertian operator $\Box : \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$ by

$$\Box \phi = \frac{\partial^2 \phi}{\partial x_1^2} - \frac{\partial^2 \phi}{\partial x_2^2} - \frac{\partial^2 \phi}{\partial x_3^2} - \frac{\partial^2 \phi}{\partial x_4^2}$$

Theorem 30.4.5 (Klein-Gordon Equations)
Let $\phi \in \mathcal{C}^\infty(M \times \mathbb{R})$. Then $\text{EL}(\phi) = 0$ if and only if

$$(\Box + m^2)\phi = 0$$

Proposition 30.4.6 (Klein-Gordon Hamiltonian)

30.5 Noether's Theorem

Definition 30.5.1 (Continuous Symmetry)

Theorem 30.5.2 (Noether's Theorem)

31 Electromagnetism

Classical field theory is developed to accommodate discoveries in the field of electromagnetism. We demonstrate the power of this language.

The Observation of Charge

Certain objects, when rubbed together, exert forces on one another across empty space. A glass rod rubbed with silk attracts a rubber rod rubbed with fur. Two glass rods rubbed with silk repel each other. Two rubber rods repel each other.

This observation suggests that matter can acquire a property which causes it to exert and respond to forces. We call this property *electric charge*. The pattern of attraction and repulsion suggests that charge comes in two varieties. We label them positive and negative, with the convention that like charges repel and opposite charges attract.

A key further observation is that charge is *conserved*: in any isolated process, the total charge never changes. When a glass rod is rubbed with silk, charge is not created — it is transferred from one object to the other, leaving the total unchanged.

Coulomb's Law and the Electric Field

In 1785, Charles-Augustin de Coulomb measured the force between charged spheres using a torsion balance. He found that the force between two point charges q_1 and q_2 separated by a distance r is:

$$F = k_e \frac{q_1 q_2}{r^2}$$

where k_e is a constant of proportionality. The force is repulsive if the charges have the same sign, and attractive if they have opposite signs. This is Coulomb's law.

The *electric field* is the conceptual tool that makes sense of how this force is transmitted. Rather than saying that charge q_1 acts directly on charge q_2 at a distance, we say that q_1 modifies the space around it, producing an electric field $\mathbb{E}$. The charge q_2 then responds to the field it finds itself in, not directly to q_1. The field is defined operationally as the force per unit charge exerted on a small positive test charge:

$$E(x) = \lim_{q \to 0} \frac{F(x)}{q}$$

The limit ensures we do not disturb the field we are trying to measure. The electric field is physically real: it carries energy and propagates at finite speed, so distant charges cannot influence each other instantaneously.

Magnetism and Faraday's Discovery

Magnetic phenomena were known long before electromagnetism was unified. Lodestones attract iron. Compass needles align with the Earth's field. For centuries, electricity and magnetism appeared to be entirely separate phenomena.

In 1820, Hans Christian Orsted discovered that a current-carrying wire deflects a compass needle placed nearby. A moving charge produces a *magnetic field* B in the space around it, just as a static charge produces an electric field. Andre-Marie Ampere then showed quantitatively that parallel currents attract, and anti-parallel currents repel.

In 1831, Michael Faraday made the reciprocal discovery: a changing magnetic field drives a current in a nearby wire. He described this in terms of *lines of force* threading through a loop. When the number of these lines changes — whether by moving a magnet, changing a current, or moving the loop itself — a voltage is induced in the loop. This is electromagnetic induction.

Faraday also observed that magnetic field lines always form closed loops. No matter how a magnet is cut or arranged, one never finds an isolated magnetic "charge" (a *monopole*) from which field lines emanate. This asymmetry between electricity and magnetism — electric charges exist, magnetic charges do not — is a fundamental observational fact.

The Unification

The deepest insight of the theory, clarified by Einstein's special relativity, is that electric and magnetic fields are not truly separate objects. They are two facets of a single *electromagnetic field*, seen from different reference frames.

An observer watching a stationary charge sees only an electric field. A second observer moving relative to the first sees the same charge in motion — a current — and therefore sees both an electric and a magnetic field. The same physical situation produces different fields depending on the observer's state of motion. This is not a paradox; it means the split between E and B is frame-dependent, while the underlying electromagnetic field is the invariant object.

Maxwell's Equations

Experiment 1: Coulomb's Law: The electric field produced by a point charge q at the origin is given by the map $E:\mathbb{R}^3\to\mathbb{R}^3$ where

$$E(x)=\frac{q}{4\pi\varepsilon_0}\frac{x}{|x|^2}$$

for some constant $\varepsilon_0\in\mathbb{R}$.

Axiom 1: Gauss's Law: Denote ρ by the charge per unit volume of $V\subseteq\mathbb{R}^3$. From the above we can deduce

$$\int_S E\cdot dS=\frac{1}{\varepsilon_0}\int_V\rho dV$$

, which equivalently says that

$$\nabla\cdot E=\frac{\rho}{\varepsilon_0}$$

Experiment 2:

$$\int_S B\cdot dS=0$$

for a closed surface S.

Axiom 2: Gauss's Law:

$$\nabla B=0$$

Experiment 3: Faraday observed

$$\int_C E\cdot dl=-\frac{d}{dt}\int_\Sigma B\cdot dS$$

where C bounds the surface Σ.

Axiom 3: Faraday's Law:

$$\nabla\times E=-\frac{\partial B}{\partial t}$$

Experiment 4: Ampere observed???

Axiom 4: Ampere Maxwell Law:

$$\nabla\times B=\mu_0J+\mu_0\varepsilon_0\frac{\partial E}{\partial t}$$

Theorem: the above implies conservation of charge.

Lorentz force law:

$$F = q(E + v \times B)$$

Something called the field potential F packs axiom 2 and 3 into a convenient package.

Notice that all the above are not time dependent. Einstein: when talking about how these fields evolve in time, the split between E and B is frame dependent, but the electromagnetic field is frame independent. (what this means: E and B come from F, F has a frame independent description, and we recover E and B by choosing a frame)

31.1 The Potential and Deriving the Classical Maxwell Equations

Definition 31.1.1 (The Four Potential)
A four potential is a smooth section $A \in \Omega^1(M)$.

We can reverse the process: the four potential gives the sourceless maxwell equations (2) and (3). Only if we assume conservation of charge, we can imply (1) and (4).

Definition 31.1.2 (The Field Tensor)
Let $A \in \Omega^1(M)$ be a four potential. Define the field tensor of A by

$$F = dA \in \Omega^2(M)$$

Definition 31.1.3 (Electric and Magnetic Fields)
Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field.

- Define the electric field of A to be

$$E = \iota_u F \in \Omega^1(M)$$

  Explicitly, we have $E_p(v) = F(u, v)$.
- Define the magnetic field of A to be

$$B = \iota_u * F \in \Omega^1(M)$$

  Explicitly, we have $B_p(v) = (*F)(u, v)$

Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding foliation (recall that they are the slices of M along a fixed time t). Then E and B restrict to each Σ_t which are the familiar electric and magnetic fields in classical electromagnetism. The restriction does not loose information because the time component is 0.

Definition 31.1.4 (Classical Electric and Magnetic Fields)
Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding inertial foliation. Let $\iota_t : \Sigma_t \hookrightarrow M$ be the inclusion.

- Define the electric field at time t to be

$$E_t = \iota_t^* E \in \Omega^1(\Sigma_t)$$

- Define the magnetic field at time t to be

$$B_t = \iota_t^* B \in \Omega^1(\Sigma_t)$$

Definition 31.1.5 (The Four Current of a Potential)
Let $A \in \Omega^1(M)$ be a four potential. Define the four current to be

$$J = *^{-1} d * F \in \Omega^1(M)$$

Definition 31.1.6 (Charge Density)
Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Define the charge density of A by
$$\rho = \iota_u J \in \mathcal{C}^\infty(M)$$

Theorem 31.1.7 (Classical Maxwell's Equations)
Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding inertial foliation. Let $\iota_t : \Sigma_t \hookrightarrow M$ be the inclusion map. Then the following are true.

- Gauss's law for electricity: $d_{\Sigma_t} *_{\Sigma_t} E_t = \iota_t^* * J$.
- Gauss's law for magnetism: $d_{\Sigma_t} B_t = 0$.
- Faraday's Law: $d_{\Sigma_t} E_t + \frac{\partial B_t}{\partial t} = 0$.
- Ampere-Maxwell Law: $d_{\Sigma_t} *_{\Sigma_t} B_t - \frac{\partial E_t}{\partial t} = \iota_t^* J_{\Sigma_t}$.

31.2 The Maxwell Lagrangian

Definition 31.2.1 (The Maxwell Lagrangian)
Define the Maxwell Lagrangian to be the map $\mathcal{L}_{\text{ML}} : J^1(T^*M) \to \text{Den}(M)$ given by
$$\mathcal{L}(j^1 A)_p = -\frac{1}{4}(dA)_p \wedge *(dA)_p$$

Proposition 31.2.2
The Euler-Lagrange operator of the Maxwell Lagrangian is given by
$$\text{EL}_{\text{ML}}(A) = d * dA - *J$$

31.3 Computing the Potential From a Current

Starting with A, we defined all the other electromagnetic quantities. Physically, we start directly with the electromagnetic field and deduce all the other quantities. On the other hand, we can start with J, and ask what the other quantities are. This is more natural because physically, starting with J is the same as specifying the matter distribution (where the charges are and how they move).

Postulate: Conservation of charge: $d * J = 0$.

Definition 31.3.1 (The Four Current)
A four current is a section $J \in \Omega^1(M)$ such that $d * J = 0$.

If we started with a four potential A, recall that J is defined by the formula $d * J = d * dA$.

Proposition 31.3.2
Let $J \in \Omega^1(M)$ be a four current. Let $A \in \Omega^1(M)$. Then $d * dA = d * J$ if and only if $\text{EL}_{\text{ML}}(A) = 0$.

The problem is that we cannot just take this as the definition of the associated potential, because A is not unique. Let $f \in \mathcal{C}^\infty(M)$. Then notice that if A is a solution, then $A + df$ is also a solution. We choose a common potential as definition (so other definitions can also possibly work, but we use the following one as the standard one).

Definition 31.3.3 (Gauge Fixing)
This is gives some boundary conditions so that the pde $d * J = d * dA$ is well posed.

The solution to the pde is given as follows:

Definition 31.3.4 (The Retarded Potential)
Let $J = \sum_{i=1}^4 J_i dx^i \in \Omega^1(M)$ be a four current. Define the retarded potential of J to be the one form $A = \sum_{i=1}^4 A_i dx^i$ whose components are given by

$$A_i(p_1, \dots, p_4) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{J_i(p_1 - \sqrt{(p_2 - x_2)^2 + (p_3 - x_3)^2 + (p_4 - x_4)^2}, x_2, x_3, x_4)}{\sqrt{(p_2 - x_2)^2 + (p_3 - x_3)^2 + (p_4 - x_4)^2}} dx_2 dx_3 dx_4$$

32 General Relativity

Equivalence Principle: Dynamics Postulate:

Minimal-coupling principle.

Definition 32.0.1 (Spacetime)
Spacetime is an n-dimensional orientable pseudo-Riemannian manifold (M, g) where g has signature $(n-1, 1)$.

Definition 32.0.2 (Einstein-Hilbert Action)

Whats next: Classical field theory (which includes relativistic field theory), continuum mechanics (which includes relativistic quantum mechanics).

33 The Logical Chain of Classical Field Theory and Gauge Theory

Step 1: The Field

We observe a physical quantity distributed over spacetime — a temperature, a velocity, an electromagnetic strength. Mathematically this is a fiber bundle $\pi : E \to M$, where M is spacetime and the fiber E_x is the space of possible values of the quantity at x. A **section** $\phi \in \Gamma(E)$ is one complete specification of the quantity at every point — a possible physical configuration.

Step 2: The Lagrangian as a Choice of Physics

A Lagrangian density $\Lambda \in \Omega^n(J^k E)$ is a choice of action functional

$$S[\phi] = \int_\Omega (j^k\phi)^*\Lambda$$

Different Lagrangians encode different physical theories. The Euler-Lagrange equations — the equations of motion — are derived from Λ by the variational principle. The physical sections are the critical points of S.

Step 3: Observing a Symmetry

You compute the Euler-Lagrange equations and notice that the Lagrangian Λ is invariant under some subgroup of automorphisms of E. An **automorphism** of $\pi : E \to M$ is a fiber-preserving diffeomorphism $\Phi : E \to E$. Such an automorphism acts on sections by

$$\phi \mapsto \Phi \circ \phi$$

and the Lagrangian being invariant means

$$S[\Phi \circ \phi] = S[\phi] \quad \text{for all } \phi \in \Gamma(E)$$

At this stage, the symmetry group acts the same way at every point — it is **global**. You observe this as an empirical fact about your Lagrangian.

Symmetries of the action induces certain laws of physics, such as conservation of energy, momentum, charge etc. This is Noether's theorem.

Another note: Rather than starting with a Lagrangian and observe the symmetries of its action, we may also start with a set of symmetries that we would like to enforce, apply it to all differential forms and retain only the invariants, and see what Lagrangian you would get.

Step 4: The Absolute vs Relative Observation

The global symmetry exists because the field takes values in a space that has **no preferred origin or frame** — the fiber E_x has a group of self-maps (rotations, phase shifts, color rotations) that are physically unobservable. Different observers, or different conventions, would describe the same physical situation using sections related by this group action. The global symmetry is an artifact of having made consistent conventional choices everywhere at once. We need not restrict ourselves to consider symmetries of the entire space, and this is supported by physics:

Argument 1: Special Relativity

Special relativity says there is no notion of simultaneity — two spacelike separated points have no canonical notion of "the same moment." A global transformation is therefore not a physically coherent operation in a relativistic theory. If you believe in special relativity, you are forced to allow your symmetry to act differently at different points. The global symmetry is then a special case of the local one, not the fundamental object.

Argument 2: The Gauge Principle

The difference induced by the symmetry of the action is not observable. The phase is therefore a **convention** — a choice made by an observer at x to describe the field. If the phase is a convention, then different observers at different points of spacetime should be free to make different conventional choices independently. There is no physical mechanism that would force their conventions to agree globally. The true symmetry group of the theory is therefore not the global group G but the local group $mathcalG = \text{Map}(M, G)$ of all smooth G-valued functions on spacetime.

This is the **gauge principle**: any redundancy in the description of a physical system should be a local redundancy. It is a postulate, not a theorem. Its justification is that it works — every time it is applied, it produces a physically correct and predictive theory.

Step 5: Promoting to Local Symmetry

This forces you to demand invariance not just under global G-transformations

$$\phi(x) \mapsto g \cdot \phi(x), \quad g \in G \text{ constant}$$

but under local G-transformations

$$\phi(x) \mapsto g(x) \cdot \phi(x), \quad g : M \to G \text{ smooth}$$

The Lagrangian, built from ordinary derivatives $\partial_\mu \phi$, immediately fails to be locally invariant. The reason is fundamental: $\partial_\mu \phi$ compares field values at nearby points x and $x + dx$, which under local symmetry live in fibers with **independent conventions**. Subtracting them is meaningless — it is comparing quantities measured in different units.

Step 6: The Mathematics Is Now Forced

To make sense of differentiation under local symmetry, we need a way to **compare vectors in nearby fibers** compatibly with the G-structure. But this is precisely the definition of a **connection**. Where does this connection live? The answer is that the principal bundle is not introduced from outside — it is **already contained inside the vector bundle** E.

The fibers E_x are vector spaces, but the isomorphism $E_x \cong \mathbb{R}^r$ is not canonical — it requires a choice of basis. A **frame** at x is an ordered basis for E_x. The set of all frames at x carries a natural free transitive right action of $GL(r, \mathbb{R})$, making it a $GL(r, \mathbb{R})$-torsor. Taking the disjoint union over all $x \in M$ produces the **frame bundle**

$$P := \bigsqcup_{x \in M} \{\text{ordered bases for } E_x\}$$

a principal $GL(r, \mathbb{R})$-bundle canonically constructed from E.

Now, the observation of a global G-symmetry in Step 3 is precisely the observation that the physically relevant frames are not arbitrary frames, but only those related by G-transformations. The structure group **reduces** from $GL(r, \mathbb{R})$ to G, producing a principal G-bundle

$$P_G \subset P$$

canonically determined by the symmetry. The original vector bundle is recovered as the associated bundle $E = P_G \times_G V$.

The connection demanded by local symmetry lives on P_G, because P_G is the bundle whose fibers are G-torsors — they encode exactly the gauge freedom at each point. A connection on P_G transports G-frames between nearby fibers, and therefore transports everything associated to those frames, including sections of $E = P_G \times_G V$.

Step 7: What Follows From The Maths

Since the principal bundle is just a collection of frames at each fiber. A local section of the principal bundle is a choice of local coordinates. The mathematical fact that non-trivial principal bundles have no global sections can be interpreted as the fact that there is generally no globally consistent choice of coordinates because the symmetries now act locally.

A connection in the principal bundle gives a covariant derivative in the associated bundle. This gives a way to compare phases across different time space.